

# 네트워크 실무

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# 1. IPTABLES 개요

# iptables

## ■ iptables 란?

- IPtables는 리눅스상에서 방화벽을 설정하는 도구
- 커널 2.4 이전 버전에서 사용되던 ipchains를 대신하는 방화벽 도구
- 커널상에서의 netfilter 패킷필터링 기능을 사용자 공간에서 제어하는 수준으로 사용

## ■ 기능

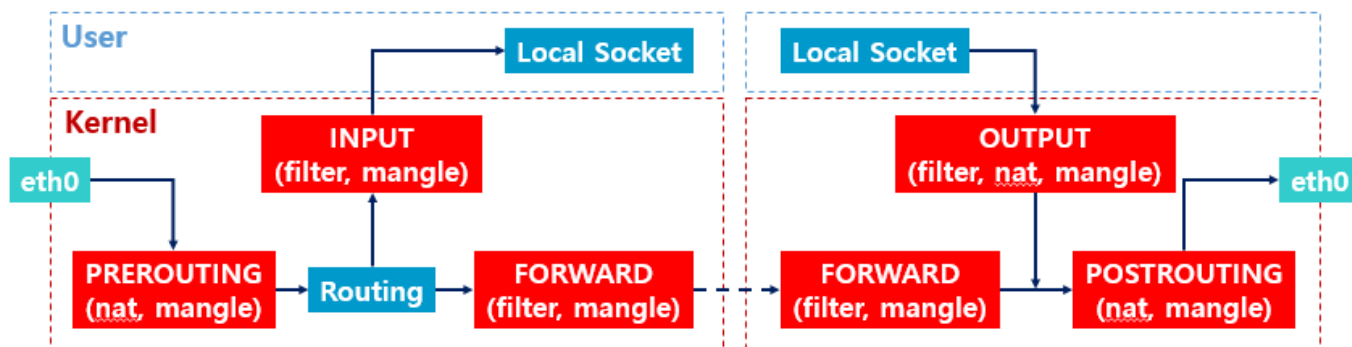
- 특정 조건을 가지고 있는 패킷에 대해 **허용(ACCEPT)**과 **차단(DROP)**등을 지정
- 특정 조건등을 통해 다양한 방식의 패킷 필터링과 처리 방식을 지원

## ■ iptables 정책

- Chain INPUT : 서버로 들어오는 기본 정책
- Chain FORWARD : 서버에서 forwarding 기본 정책
- Chain OUTPUT : 서버에서 나가는 기본 정책

# iptables

## ■ iptables 동작



- Linux Server를 목적지로 삼는 모든 패킷은 INPUT Chain을 통과
- Linux Server에서 생성되어 외부로 보내지는 모든 패킷은 OUTPUT Chain을 통과
- FORWARD Chain의 경우 현재의 Linux Server가 목적지가 아닌 패킷이 통과하는 Chain (FORWARD Chain은 NAT(네트워크 공유) 기능 사용을 위해 사용)

## ■ 마스커레이드(Masquerade)

- 내부 사설 IP의 PC들이 외부 인터넷이 연결 가능하도록 해주는 기능

## ■ NAT(Network Address Translation) : 네트워크 주소 변환 서비스

- SNAT(Source NAT) : 내부 사설IP에서 외부로 나갈 때 공인IP로 변환 -> 마스커레이드와 비슷
- DNAT(Destination NAT) : 외부에서 방화벽(외부IP)으로 요청되는 주소로 내부사설IP로 변환

# iptables의 구조

## ■ 테이블 (Table)

- iptables는 체인이라는 규칙을 연결한 테이블 구조로 되어 있다.
- 각 테이블은 자신만의 고유한 사슬 집합을 가지고 있지만, 사용자가 공통태그와 관련된 규칙집합을 만들기 위해 사용자 정의 사슬을 생성할 수 있다
- iptables 는 filter, nat, mangle, raw, security와 같은 5개의 테이블
  - filter : iptables의 기본 테이블로 패킷 필터링 담당
  - nat : Network Address Translation, IP 주소 변환
  - mangle : 패킷 데이터를 변경하는 특수 규칙을 적용하는 테이블, 성능향상을 위한 TOS(Type of Service) 설정
  - raw : 넷필터의 연결추적 하위시스템과 독립적으로 동작해야 하는 규칙을 설정하는 테이블
  - security : 리눅스 보안 모듈인 SELinux에 의해 사용되는 MAC(Mandatory Access Control) 네트워크 관련 규칙 적용

# iptables의 구조

## ■ 체인 (Chain)

- 일치하는 패킷의 규칙을 연결하는 단위
- INPUT, OUTPUT, FORWARD가 내장 체인으로 되어있다
  - INPUT : 시스템 외부에서 내부로 들어오는 체인
  - OUTPUT : 시스템 내부에서 외부로 나가는 체인
  - FORWARD : 시스템을 통과하는 체인

# iptables 사용법

- iptables [-t table] [action] [chain] [match] [-j target]
  - iptables를 이용하여 정책을 설정할 때 가장 중요한 것은 실질적 룰(Rule)에 해당하는 매치(match)와 타깃(Target)이다.
  - 타깃 : iptables에서 패킷이 규칙과 일치 할 때 취하는 동작
  - 매치 : iptables가 규칙 타깃에 의해 명시되는 동작에 따라 패킷을 처리하기 위해 만족해야 하는 조건들

# iptables 사용법

## ■ 주요 action

| action | 설명                                     |
|--------|----------------------------------------|
| -N     | 새로운 사용자 정의 사슬                          |
| -X     | 비어있는 사슬 제거, 기본 사슬 제거불가                 |
| -P     | 사슬의 기본 정책 설정 (policy)                  |
| -L     | 현재 사슬의 규칙 나열                           |
| -F     | 사슬로부터 규칙 전부 제거 (flush)                 |
| -Z     | 사슬내의 모든 규칙들의 패킷과 바이트의 카운트를 0으로 만든다     |
| -A     | 사슬에 새로운 규칙 추가, 맨 마지막 규칙으로 등록됨 (append) |
| -I     | 사슬에 규칙을 맨 첫 부분에 삽입                     |
| -D     | 사슬의 규칙을 제거                             |



# iptables 사용법

## ■ 주요 match

| match | 설명                                                                                        |
|-------|-------------------------------------------------------------------------------------------|
| -s    | 출발지 IP주소나 네트워크와 매칭, 도메인, IP주소, 넷마스크값 이용하여 표시                                              |
| -d    | 목적지 IP주소나 네트워크와 매칭, 도메인, IP주소, 넷마스크값 이용하여 표시                                              |
| -p    | 특정 프로토콜과 매칭, TCP, UDP, ICMP 와 같은 이름을 사용하고 대소문자는 구분하지<br>않음<br>이 옵션 사용하지 않으면 모든 프로토콜이 대상이됨 |
| -i    | 입력 인터페이스와 매칭                                                                              |
| -o    | 출력 인터페이스와 매칭                                                                              |

## ■ iptables -p 옵션 뒤에 프로토콜 식명 추가 옵션

| option  | 설명                                   |
|---------|--------------------------------------|
| --sport | 발신지에서의 하나의 포트 또는 포트범위를 지정            |
| --dport | 도착지의 포트를 지정하는 것으로 설정 방법은 --sport와 동일 |

# iptables 사용법

## ■ 주요 target (-j target)

| target | 설명                                                |
|--------|---------------------------------------------------|
| ACCEPT | 패킷을 허가하는 것으로 본래 라우팅대로 진행                          |
| DROP   | 패킷을 거부하는 것으로 더 이상 어떤 처리도 수행하지 않고 버림               |
| LOG    | 패킷을 syslog에 전달하여 기록, 일반적으로 /var/log/messages 에 저장 |
| REJECT | 패킷을 버리고 동시에 적당한 응답 패킷을 전달                         |
| RETURN | 호출 사슬 내에서 패킷 처리를 계속 진행                            |

```
# iptables -F
-> 현재 설정되어 있는 iptables의 규칙(rule)을 전부 제거
# iptables -P INPUT DROP
-> 들어오는 모든 패킷을 거부한다
# iptables -A INPUT -s 192.168.5.13 -j ACCEPT
-> IP주소가 192.168.5.13인 호스트로부터 들어오는 패킷을 허가한다

# iptables -F INPUT
# iptables -A INPUT -p tcp --dport 22 -j ACCEPT
# iptables -A INPUT -p tcp --dport 20:30 -j DROP
-> 포트번호 20 - 30번까지 차단하고, ssh 포트인 22번만 허가한다
-> 거부 메시지 없이 무조건 거절한다
```

# 방화벽 규칙(Rule)을 확인하고 저장 및 관리

```
# iptables-save > firewall.sh
```

-> 현재 설정되어 있는 iptables의 규칙(rule)을 firewall.sh 파일로 저장한다.

```
# iptables -F
```

-> 현재 설정되어 있는 iptables의 규칙(rule)을 전부 제거한다.

```
# iptables-restore < firewall.sh
```

-> firewall.sh 파일에 저장되어 있는 iptables의 규칙(rule)을 불러와서(복구) 반영한다.

# iptables 규칙

## ■ iptables 규칙 추가

- 127.0.0.1 즉, 로컬에서 요청하는 모든 ICMP 패킷에 대해 무시하고자 할 때
- ping 프로토콜은 ICMP이며, 발신 주소는 127.0.0.1, 패킷 필터의 목표는 폐기(DROP)
  - # iptables -A INPUT -s 127.0.0.1 -p icmp -j DROP

## ■ iptables 규칙 제거

- iptables의 체인 입력과 마찬가지로 삭제를 할 때에도 동일하게 입력하면 된다.
- iptables -D INPUT -s [발신지] --sport [발신지 포트] -d [목적지] --dport [목적지 포트] -j [정책]
- iptables -D INPUT [필터링 번호]  
(필터링 번호는 service iptables stat을 통해 확인할 수 있다.)
- iptables -F INPUT 을 입력할 경우 모든 체인이 삭제된다.
- 위에 입력했던 체인을 제거
  - # iptables -D INPUT 1 또는
  - # iptables -D INPUT -s 127.0.0.1 -p icmp -j DROP

# iptables 정책 순서

## ■ 모든 방화벽은 순차적 실행

- 먼저 등록한 부분에 대해서 효력이 유효하기 때문에 등록시에는 순서가 매우 중요
- 허용하는 정책을 먼저 정의한 다음 거부하는 정책을 설정
- 다음같이 설정 시 22번 포트가 열린 후 나중에 22번~30번 포트가 막히기 때문에 SSH 접속이 가능
  - # iptables -A INPUT -p tcp --dport 22 -j ACCEPT
  - # iptables -A INPUT -p tcp --dport 22:30 -j DROP
- 다음같이 설정 시 22번~30번 포트가 막히기 때문에 뒤에서 아무리 22번 포트를 열어도 외부에서 SSH로 접속 불가
  - # iptables -A INPUT -p tcp --dport 22:30 -j DROP
  - # iptables -A INPUT -p tcp --dport 22 -j ACCEPT

# iptables 활용 예

## ■ 활용 예

- 소스 ip가 192.168.0.111 인 접속의 모든 접속 포트를 막아라.
  - # iptables -A INPUT -s 192.168.0.111 -j DROP
- INPUT 사슬에 출발지 주소가 127.0.0.1(-s 127.0.0.1) 인 icmp 프로토콜(-p icmp) 패킷을 거부(-j DROP)하는 정책을 추가(-A)하라
  - # iptables -A INPUT -p icmp -s 127.0.0.1 -j DROP
- INPUT 사슬에 목적지 포트가 23번(--dport 23)인 tcp 프로토콜(-p tcp) 패킷을 거부하는(-j DROP)규칙을 추가(-A) 하라.
  - # iptables -A INPUT -p tcp --dport 23 -j DROP
- INPUT 사슬에 목적지 포트 번호가 80번(--dport 80)인 tcp 프로토콜(-p tcp)패킷을 받아들이는(-j ACCEPT) 규칙을 추가(-A) 하라
  - # iptables -A INPUT -p tcp --dport 80 -j ACCEPT
- INPUT 사슬에 목적지 포트번호가 1023번 보다 작은 모든 포트(--dport :1023)인 tcp 프로토콜(-p tcp)패킷을 거부하는(-j DROP)규칙을 추가(-A)하라
  - # iptables -A INPUT -p tcp --dport :1023 -j DROP

# iptables 활용 예

## ■ 활용 예

### ➤ ftp포트를 열어라

- # iptables -I INPUT -p tcp --dport 21 -j ACCEPT

### ➤ imap 서비스를 방화벽에서 열어라

- # iptables -I INPUT -s 192.168.0.0/255.255.255.0 -p udp --dport 143 -j ACCEPT

### ➤ 웹서버 서비스를 방화벽에서 열어라

- # iptables -I INPUT -p tcp --dport 80 -j ACCEPT

### ➤ 웹서버 포트 80 -> 8880으로 교체하라( 웹서비스 포트 변경시 /etc/services 에서도 변경 해줘야 함)

- # iptables -R INPUT 2 -p tcp --dport 8880 -j ACCEPT

### ➤ domain-access\_log 파일에 있는 모든 ip의 모든 접속 포트를 막아라(DOS공격 방어 시 사용)

- # cat domain-access\_log |awk '{print \$1}'|sort |uniq |awk '{print "iptables -A INPUT -s \"\$1\" -j DROP"}'|/bin/bash

# 전체 방화벽 설정 초기화

- 이전 방화벽 정책들이 존재하는 경우 모든 내용을 초기화
  - iptables -P INPUT ACCEPT
  - iptables -P OUTPUT ACCEPT
  - iptables -P FORWARD ACCEPT
  - iptables -F
- 방화벽 서비스 부팅
  - 서비스 시작: service iptables start
  - 서비스 정지: service iptables stop
  - 서비스 재시작: service iptables restart
  - 서비스 저장: service iptables save



# iptables를 안정적으로 사용하기 위한 패키지 설치

- iptables은 시스템을 다시 시작하거나, 서비스를 다시 시작하면 설정이 초기화
- 최신 버전의 우분투(Ubuntu)에서 빈번히 나타는 현상이므로 이를 해소할 수 있는 패키지 설치
  - apt-get install iptables-persistent netfilter-persistent
- 설치 도중에 ip4/ip6 저장 여부를 묻는데 yes를 선택

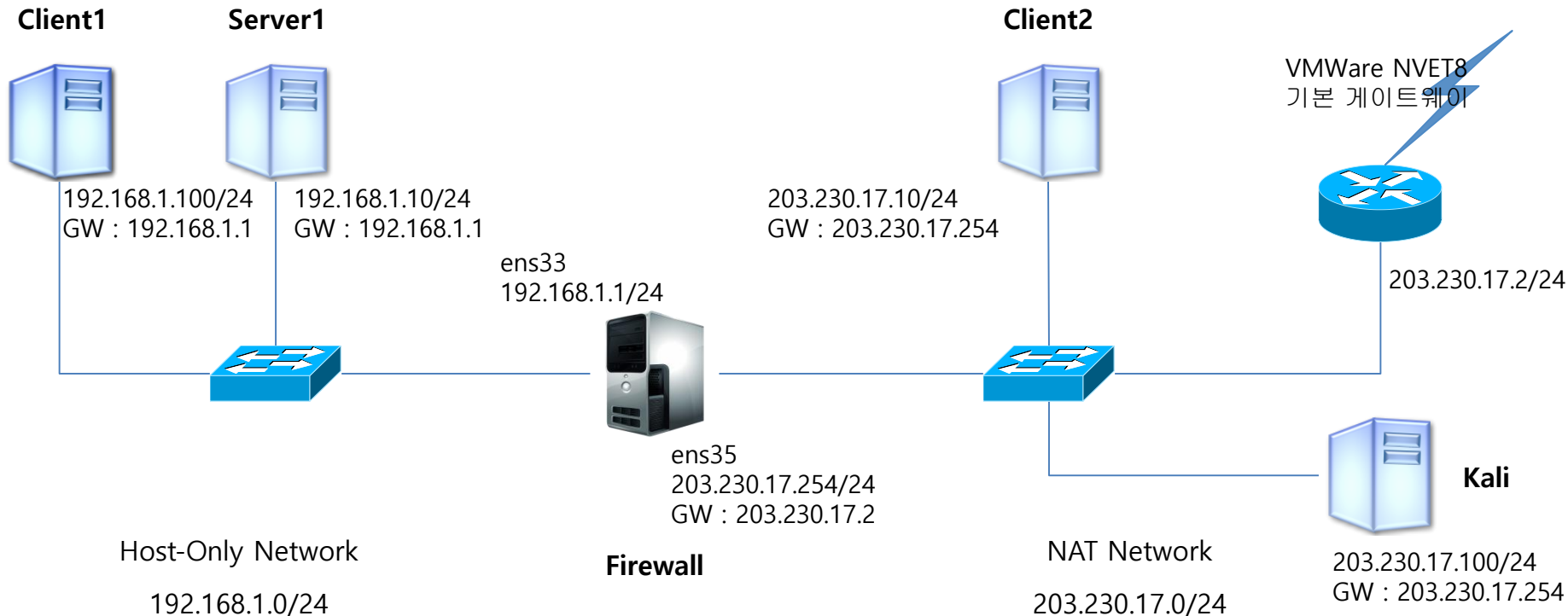
```
| Configuring iptables-persistent |  
  
Current iptables rules can be saved to the configuration file /etc/iptables/rules.v4. These rules  
will then be loaded automatically during system startup.  
  
Rules are only saved automatically during package installation. See the manual page of  
iptables-save(8) for instructions on keeping the rules file up-to-date.  
  
Save current IPv4 rules?  
  
      <Yes>                                <No>
```

- 설치를 마치고나면 /etc/iptables 라는 폴더에 rules.v4와 rules.v6가 생성
- iptables 설정을 변경 후 netfilter-persistent save 명령어로 저장
- netfilter-persistent reload 명령어로 다시 설정을 불러올 수 있음
  - sudo netfilter-persistent save
  - sudo netfilter-persistent reload

## 2. LINUX ACL

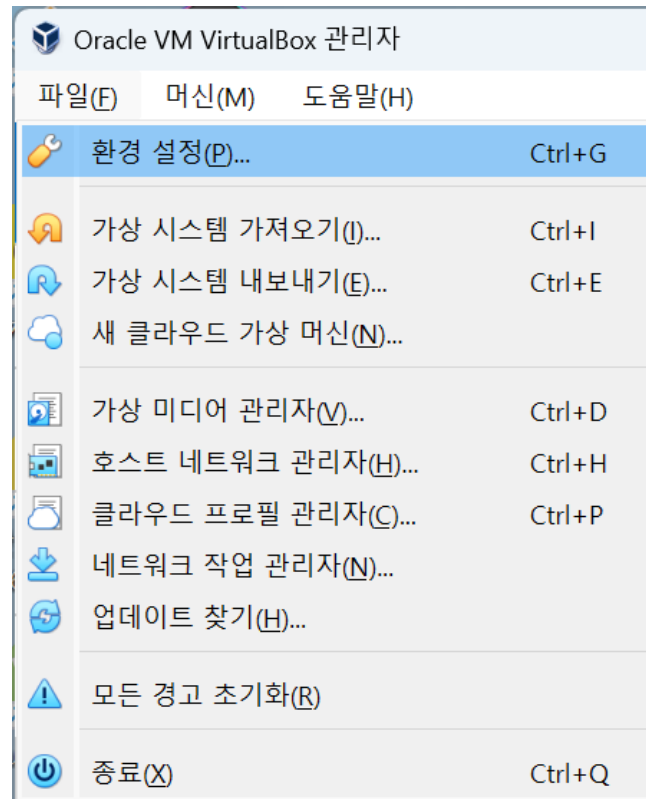
# 구성도

- 우분투 리눅스로 방화벽을 만들어 서로 다른 네트워크간 ACL 통신을 할 수 있도록 설정하도록 합니다.



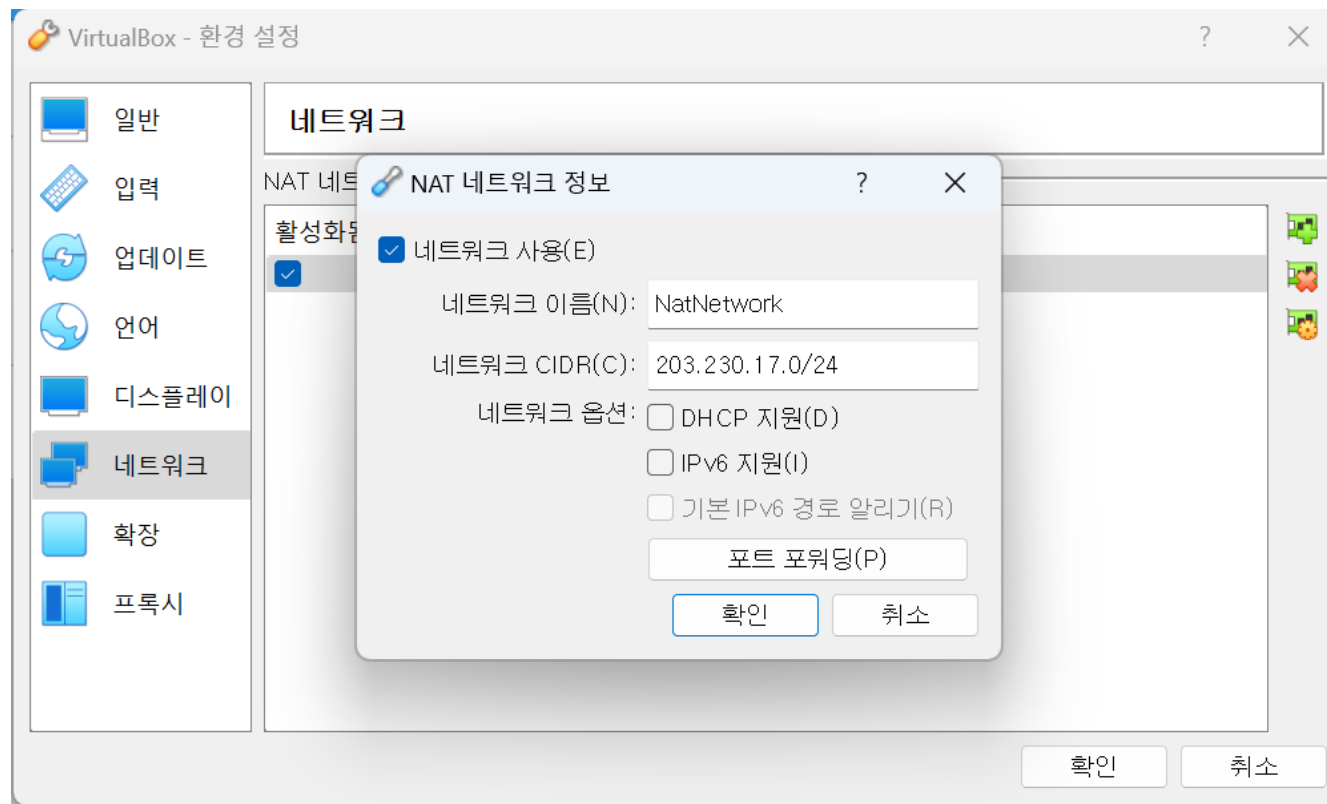
# Network 환경 설정

- VirtualBox 프로그램에서 '파일(F)' → '환경설정(P)...'를 실행한다.



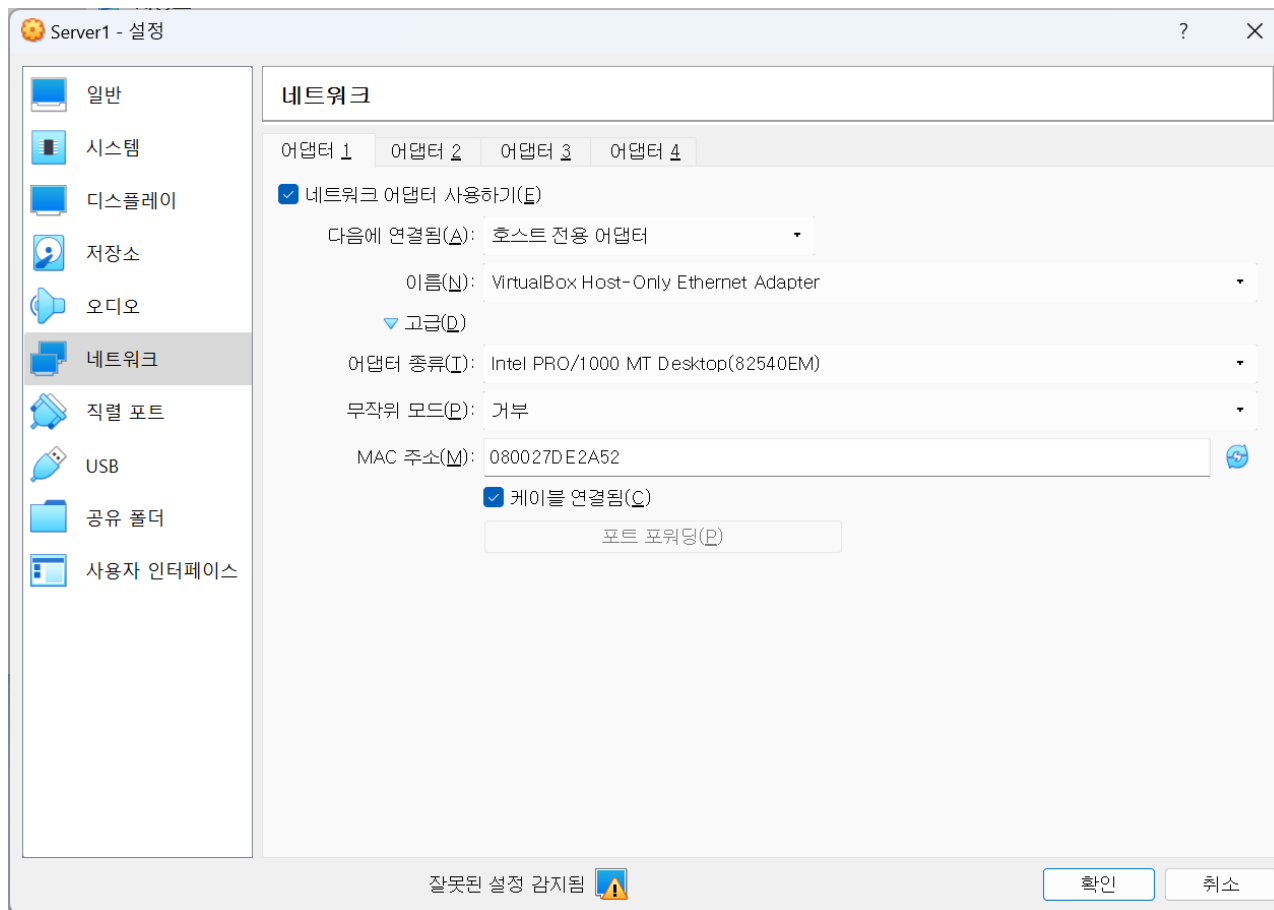
# Network 환경 설정

- 네트워크를 선택하고 NAT 네트워크를 추가한다.
  - 네트워크 CIDR© : 203.230.17.0/24
  - DHCP지원 → 제거



# Server1 설정

## ■ Network Adapter : VirtualBox Host-Only Ethernet Adapter



# Server1 설정

## ■ Ubuntu Network 설정

- IPv4 Method : Manual
- IP : 192.168.1.10
- Netmask : 24
- Gateway : 192.168.1.1
- DNS : 8.8.8.8

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

**IPv4 Method**

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

**Addresses**

| Address      | Netmask       | Gateway     |  |
|--------------|---------------|-------------|--|
| 192.168.1.10 | 255.255.255.0 | 192.168.1.1 |  |
|              |               |             |  |

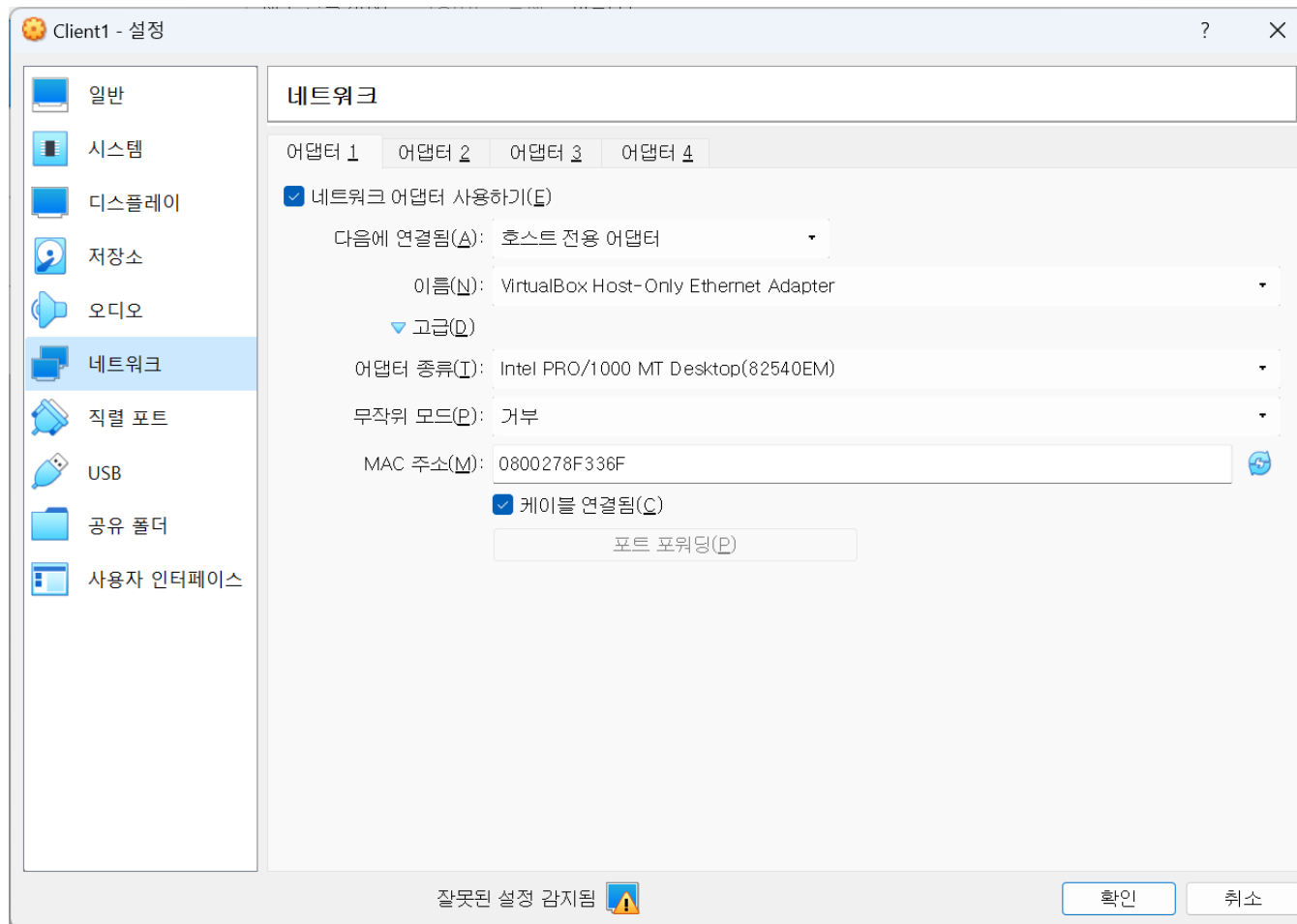
**DNS** Automatic ☒

8.8.8.8

Separate IP addresses with commas

# Client1 설정

- Network Adapter : VirtualBox Host-Only Ethernet Adapter





# Client1 설정

## ■ Ubuntu Network 설정

- IPv4 Method : Manual
- IP : 192.168.1.100
- Netmask : 24
- Gateway : 192.168.1.1
- DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

**IPv4 Method**

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

**Addresses**

| Address       | Netmask       | Gateway     |  |
|---------------|---------------|-------------|--|
| 192.168.1.100 | 255.255.255.0 | 192.168.1.1 |  |
|               |               |             |  |

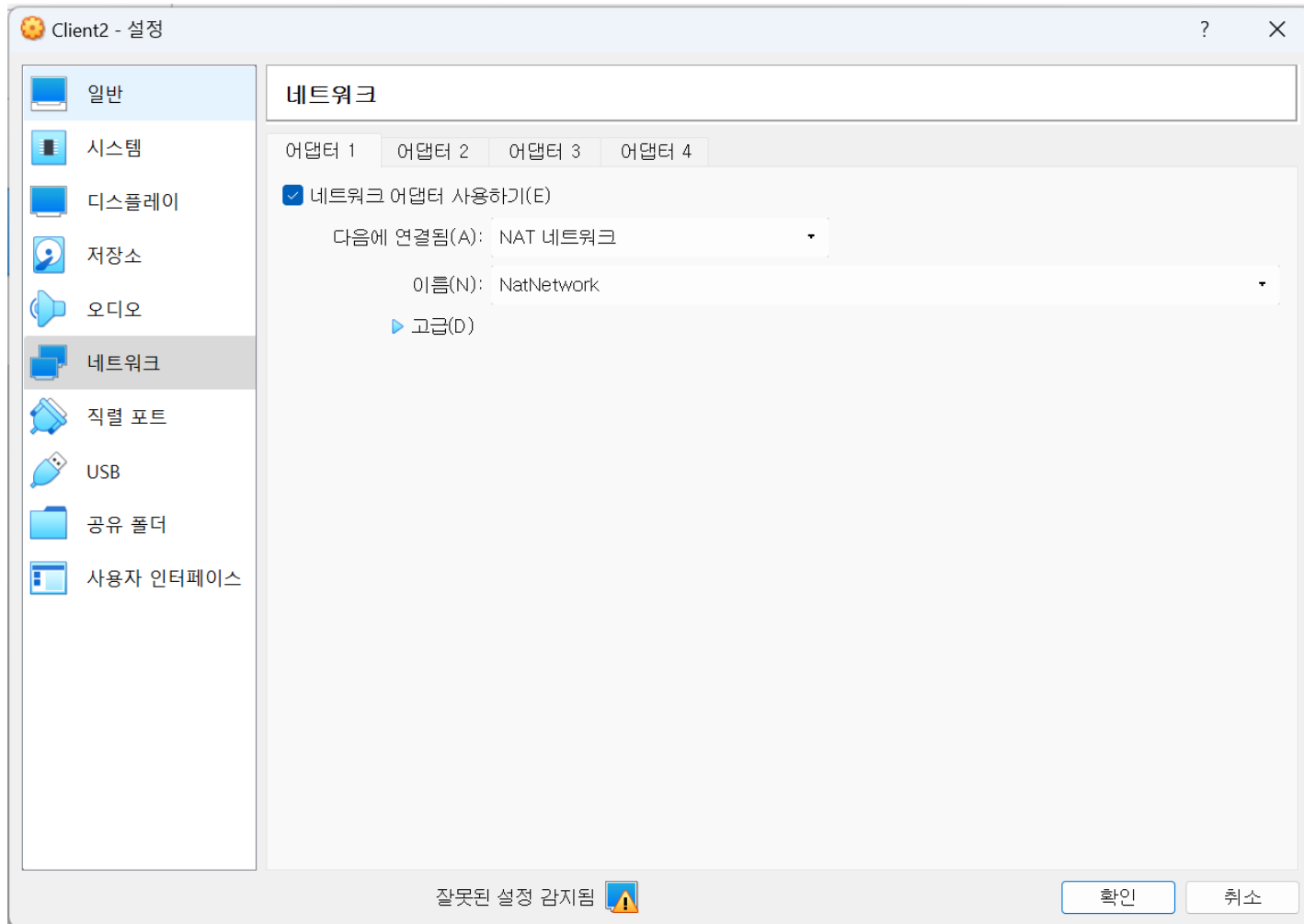
**DNS** Automatic ☒

8.8.8.8

Separate IP addresses with commas

# Client2 설정

- Network Adapter : NAT 네트워크 -> NATNetwork 선택



# Client2 설정

## ■ Ubuntu Network 설정

- IPv4 Method : Manual
- IP : 203.230.17.10
- Netmask : 24
- Gateway : 203.230.17.254
- DNS : 8.8.8.8

Cancel Wired Apply

Details Identity **IPv4** IPv6 Security

**IPv4 Method**

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

**Addresses**

| Address       | Netmask       | Gateway        |  |
|---------------|---------------|----------------|--|
| 203.230.17.10 | 255.255.255.0 | 203.230.17.254 |  |
|               |               |                |  |

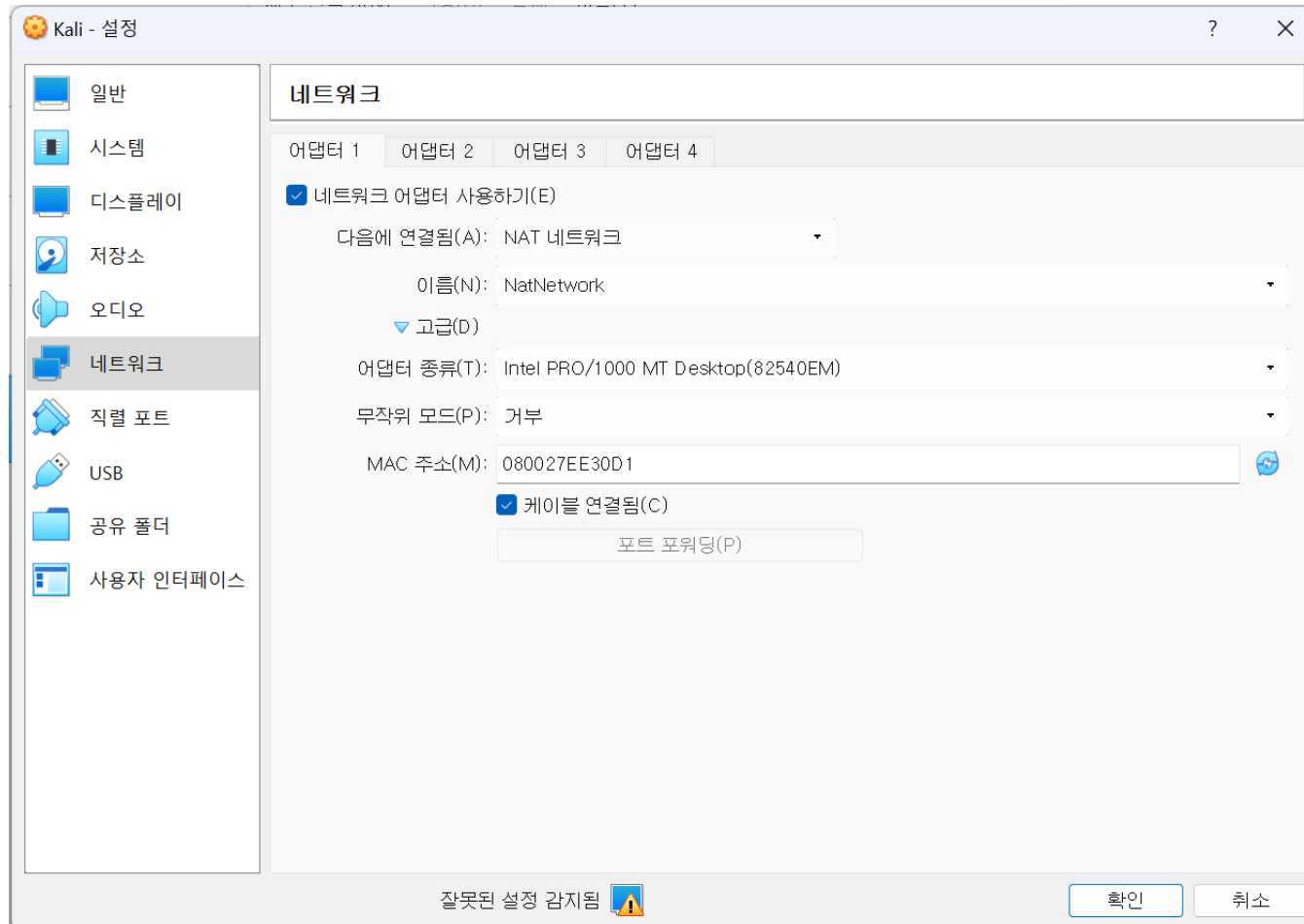
**DNS** Automatic ☒

8.8.8.8

Separate IP addresses with commas

# Kali 설정

- Network Adapter : NAT 네트워크 -> NATNetwork 선택



# Kali 설정

## ■ Kali Network 설정

- IPv4 Method : Manual
- IP : 203.230.17.100
- Netmask : 24
- Gateway : 203.230.17.254
- DNS : 8.8.8.8

Editing Wired connection 1

Connection name: **Wired connection 1**

General Ethernet 802.1X Security DCB Proxy **IPv4 Settings** IPv6 Settings

Method: **Manual**

Addresses

| Address        | Netmask | Gateway        |
|----------------|---------|----------------|
| 203.230.17.100 | 24      | 203.230.17.254 |

DNS servers: **8.8.8.8**

Search domains:

DHCP client ID:

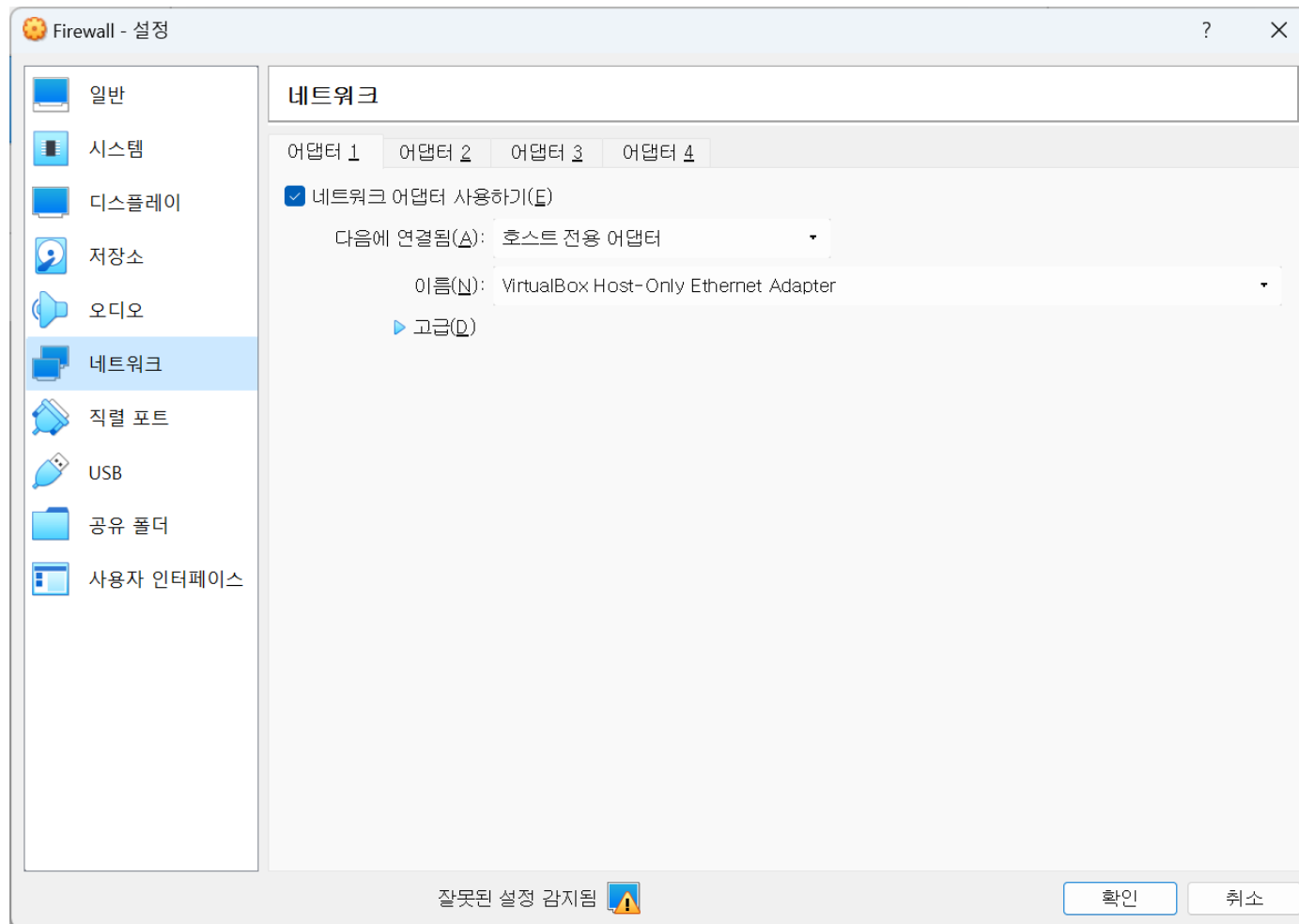
☐ Require IPv4 addressing for this connection to complete

Routes...

Cancel Save

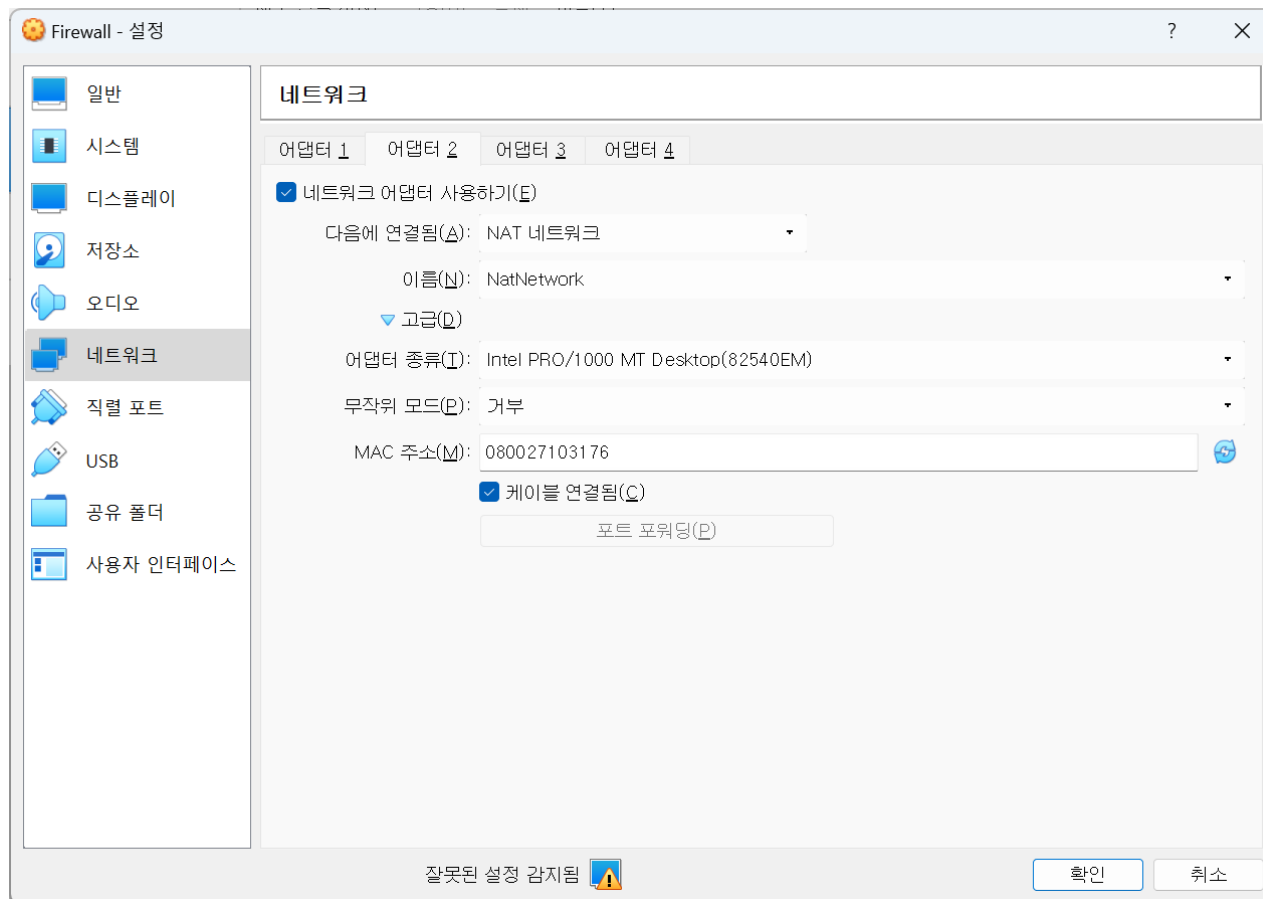
# Firewall 설정

- Network Adapter : VirtualBox Host-Only Ethernet Adapter



# Firewall 설정

- Network Adapter 2 : NAT 네트워크 -> NATNetwork 선택



# Firewall 설정

- Ubuntu Network 설정
  - Interface : enp0s3
  - IPv4 Method : Manual
  - IP : 192.168.1.1
  - Netmask : 24
  - DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

**IPv4 Method**

☐ Automatic (DHCP) ☐ Link-Local Only

☒ **Manual** ☐ Disable

☐ Shared to other computers

**Addresses**

| Address     | Netmask       | Gateway |  |
|-------------|---------------|---------|--|
| 192.168.1.1 | 255.255.255.0 |         |  |
|             |               |         |  |

**DNS** Automatic ☒

8.8.8.8

Separate IP addresses with commas



# Firewall 설정

## ■ Ubuntu Network 2 설정

- Interface : enp0s8
- IPv4 Method : Manual
- IP : 203.230.17.254
- Netmask : 24
- Gateway : 203.230.17.2
- DNS : 8.8.8.8

Cancel **Wired** Apply

Details Identity **IPv4** IPv6 Security

**IPv4 Method**

☐ Automatic (DHCP) ☐ Link-Local Only

☒ Manual ☐ Disable

☐ Shared to other computers

**Addresses**

| Address        | Netmask       | Gateway      |  |
|----------------|---------------|--------------|--|
| 203.230.17.254 | 255.255.255.0 | 203.230.17.2 |  |
|                |               |              |  |

**DNS** Automatic ☒

8.8.8.8

Separate IP addresses with commas

# Firewall 설정

## ■ Interface간 패킷 전달

➤ enp0s3 Interface와 enp0s85 Interface간 패킷 전송을 위한 Routing 기능 활성화

➤ \$ sudo vi /etc/sysctl.conf

```
# Uncomment the next two lines to enable Spoof protection (reverse-path filter)
# Turn on Source Address Verification in all interfaces to
# prevent some spoofing attacks
#net.ipv4.conf.default.rp_filter=1
#net.ipv4.conf.all.rp_filter=1

# Uncomment the next line to enable TCP/IP SYN cookies
# See http://lwn.net/Articles/277146/
# Note: This may impact IPv6 TCP sessions too
#net.ipv4.tcp_syncookies=1

# Uncomment the next line to enable packet forwarding for IPv4
net.ipv4.ip_forward=0

# Uncomment the next line to enable packet forwarding for IPv6
# Enabling this option disables Stateless Address Autoconfiguration
# based on Router Advertisements for this host
#net.ipv6.conf.all.forwarding=1
```

➤ #net.ipv4.ip\_forwarding=1 행을 찾아서 맨 앞의 # 제거 후 저장

➤ 변경된 내용을 적용

➤ \$ sudo sysctl -p /etc/sysctl.conf

```
ubuntu@Firewall:~/Desktop$ sudo sysctl -p /etc/sysctl.conf
[sudo] password for ubuntu:
net.ipv4.ip_forward = 1
ubuntu@Firewall:~/Desktop$
```

# 라우팅 통신 확인

## ■ 방화벽 기준 내외부 통신 테스트

### ➤ 방화벽 정책 확인

```
root@Firewall: ~  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

### ➤ Kali 에서 Server1 통신 테스트

```
root@kali: /home/kali  
File Actions Edit View Help  
  
(root@kali)-[/home/kali]  
# ping 192.168.1.10 -c 2  
PING 192.168.1.10 (192.168.1.10) 56(84) bytes of data.  
64 bytes from 192.168.1.10: icmp_seq=1 ttl=63 time=0.698 ms  
64 bytes from 192.168.1.10: icmp_seq=2 ttl=63 time=0.249 ms  
  
— 192.168.1.10 ping statistics —  
2 packets transmitted, 2 received, 0% packet loss, time 1005ms  
rtt min/avg/max/mdev = 0.249/0.473/0.698/0.224 ms  
  
(root@kali)-[/home/kali]  
#
```

# 라우팅 통신 확인

## ■ Client1 → Client2 통신 테스트

### ➤ SSH 통신

```
ubuntu@client2: ~  
root@client1:~# ssh ubuntu@203.230.17.10  
The authenticity of host '203.230.17.10 (203.230.17.10)' can't be established.  
ECDSA key fingerprint is SHA256:ctYRFLTJXdebHZAe8G1097tfE050GhhUrL2I3Z5t5Xk.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '203.230.17.10' (ECDSA) to the list of known hosts.  
ubuntu@203.230.17.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
Your Hardware Enablement Stack (HWE) is supported until 2025-04-01.  
*** System restart required ***  
  
The programs included with the Ubuntu system are free of charge;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
  
ubuntu@client2:~$
```

```
ubuntu@client1: ~/Desktop  
ubuntu@client1:~/Desktop$ netstat -ltn  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 192.168.1.100:47204     203.230.17.10:22       ESTABLISHED  
tcp        0      1 192.168.1.100:45516     8.8.8.8:53              SYN_SENT  
tcp6       0      0 :::22                   :::*                    LISTEN  
tcp6       0      0 :::1:631                :::*                    LISTEN  
ubuntu@client1:~/Desktop$
```

```
root@client2: ~  
root@client2:~# netstat -ltn  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 203.230.17.10:22        192.168.1.100:47204     ESTABLISHED  
tcp6       0      0 :::22                   :::*                    LISTEN  
tcp6       0      0 :::1:631                :::*                    LISTEN  
root@client2:~#
```

# 라우팅 통신 확인

## ■ Client2 → Server1 통신 테스트

### ➤ SSH 통신

```
ubuntu@Server1: ~  
root@client2:~# ssh ubuntu@192.168.1.10  
ubuntu@192.168.1.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Failed to connect to https://changelogs.ubuntu.com/  
Internet connection or proxy settings  
  
Your Hardware Enablement Stack (HWE) is supported  
Last login: Sun Nov 27 19:52:00 2022 from 203.230.1  
ubuntu@Server1:~$
```

```
root@Server1: ~  
root@Server1:~# netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 192.168.1.10:22        203.230.17.10:34648     ESTABLISHED  
tcp        0      1 192.168.1.10:56520      8.8.8.8:53              SYN_SENT  
tcp6       0      0 :::22                  :::*                   LISTEN  
tcp6       0      0 :::1:631                :::*                   LISTEN
```

```
ubuntu@client2: ~  
ubuntu@client2:~/Desktop$ cd  
ubuntu@client2:~$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 203.230.17.10:36020     91.189.91.49:80        TIME_WAIT  
tcp        0      0 203.230.17.10:34648     192.168.1.10:22        ESTABLISHED  
tcp6       0      0 :::22                  :::*                   LISTEN  
tcp6       0      0 :::1:631                :::*                   LISTEN
```

# ACL 통신 확인(서버)

## ■ Server1 ACL 정책 적용

### ➤ 정책 확인

```
root@Server1: ~  
root@Server1:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Server1:~#
```

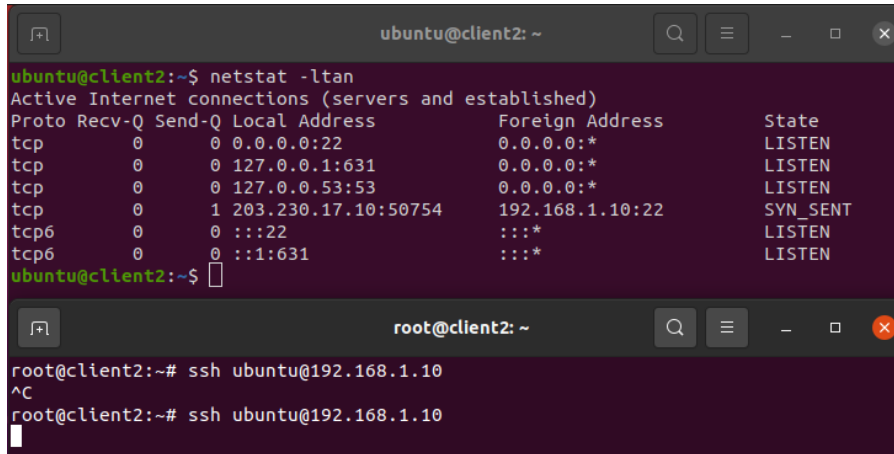
### ➤ 192.168.1.0/24에서만 ssh접속 가능 정책 설정

```
root@Server1: ~  
root@Server1:~# iptables -I INPUT -s 192.168.1.0/255.255.255.0 -p tcp --dport 22 -j ACCEPT  
root@Server1:~# iptables -A INPUT -p tcp --dport 22 -j DROP  
root@Server1:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
ACCEPT      tcp  --  192.168.1.0/24         anywhere             tcp dpt:ssh  
DROP        tcp  --  anywhere              anywhere             tcp dpt:ssh  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Server1:~#
```

# ACL 통신 확인(서버)

## ■ Client2 → Server1 통신 테스트(비정상 통신)

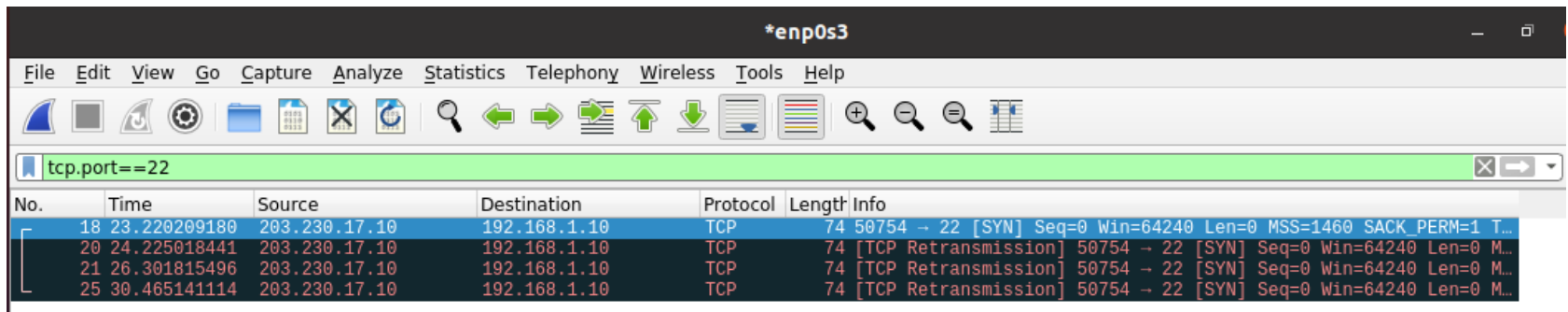
### ➤ SSH 통신 시도



```
ubuntu@client2:~$ netstat -ltan
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN
tcp        0      1 203.230.17.10:50754     192.168.1.10:22        SYN_SENT
tcp6       0      0 :::22                  :::*                    LISTEN
tcp6       0      0 :::1:631                :::*                    LISTEN

ubuntu@client2:~$
root@client2:~# ssh ubuntu@192.168.1.10
^C
root@client2:~# ssh ubuntu@192.168.1.10
```

### ➤ Client1 에서의 패킷



Filter: tcp.port==22

| No. | Time         | Source        | Destination  | Protocol | Length | Info                                                             |
|-----|--------------|---------------|--------------|----------|--------|------------------------------------------------------------------|
| 18  | 23.220209180 | 203.230.17.10 | 192.168.1.10 | TCP      | 74     | 50754 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T... |
| 20  | 24.225018441 | 203.230.17.10 | 192.168.1.10 | TCP      | 74     | [TCP Retransmission] 50754 → 22 [SYN] Seq=0 Win=64240 Len=0 M... |
| 21  | 26.301815496 | 203.230.17.10 | 192.168.1.10 | TCP      | 74     | [TCP Retransmission] 50754 → 22 [SYN] Seq=0 Win=64240 Len=0 M... |
| 25  | 30.465141114 | 203.230.17.10 | 192.168.1.10 | TCP      | 74     | [TCP Retransmission] 50754 → 22 [SYN] Seq=0 Win=64240 Len=0 M... |

# ACL 통신 확인(서버)

## ■ Client1 → Server1 통신 테스트

### ➤ SSH 통신 시도

```
ubuntu@Server1: ~  
root@client1:~# ssh ubuntu@192.168.1.10  
The authenticity of host '192.168.1.10 (192.168.1.10)' can't be established.  
ECDSA key fingerprint is SHA256:ctYRFLTJXdebHZAe8G1097tFE050GhhUeL2I3Z5t5Xk.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.1.10' (ECDSA) to the list of known hosts  
ubuntu@192.168.1.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your  
Internet connection or proxy settings  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
Last login: Sun Nov 27 19:52:10 2022 from 203.230.17.10  
ubuntu@Server1:~$
```

```
ubuntu@client1: ~  
ubuntu@client1:~$ netstat -ltn  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 192.168.1.100:42452     192.168.1.10:22        ESTABLISHED  
tcp        0      1 192.168.1.100:35618     8.8.8.8:53             SYN_SENT  
tcp6       0      0 :::22                   :::*                    LISTEN  
tcp6       0      0 :::1:631                :::*                    LISTEN
```

### ➤ Client1 에서의 패킷

| No. | Time         | Source        | Destination  | Protocol | Length | Info                                                             |
|-----|--------------|---------------|--------------|----------|--------|------------------------------------------------------------------|
| 21  | 19.324945970 | 192.168.1.100 | 192.168.1.10 | TCP      | 74     | 42452 → 22 [SYN] Seq=0 Win=64240 Len=0 MSS=1460 SACK_PERM=1 T... |
| 22  | 19.324962515 | 192.168.1.100 | 192.168.1.10 | TCP      | 74     | 22 → 42452 [SYN, ACK] Seq=0 Ack=1 Win=65160 Len=0 MSS=1460 SA... |
| 23  | 19.325138644 | 192.168.1.100 | 192.168.1.10 | TCP      | 66     | 42452 → 22 [ACK] Seq=1 Ack=1 Win=64256 Len=0 TSval=3718391413... |
| 24  | 19.325230089 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 107    | Client: Protocol (SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5)       |
| 25  | 19.325235295 | 192.168.1.100 | 192.168.1.10 | TCP      | 66     | 22 → 42452 [ACK] Seq=1 Ack=42 Win=65152 Len=0 TSval=714565316... |
| 26  | 19.329136340 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 107    | Server: Protocol (SSH-2.0-OpenSSH_8.2p1 Ubuntu-4ubuntu0.5)       |
| 27  | 19.329384717 | 192.168.1.100 | 192.168.1.10 | TCP      | 66     | 42452 → 22 [ACK] Seq=42 Ack=42 Win=64256 Len=0 TSval=37183914... |
| 28  | 19.329470234 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 1578   | Client: Key Exchange Init                                        |
| 29  | 19.329787024 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 1122   | Server: Key Exchange Init                                        |
| 30  | 19.330902071 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 114    | Client: Diffie-Hellman Key Exchange Init                         |
| 31  | 19.332904852 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 574    | Server: Diffie-Hellman Key Exchange Reply, New Keys, Encrypte... |
| 32  | 19.375340780 | 192.168.1.100 | 192.168.1.10 | TCP      | 66     | 42452 → 22 [ACK] Seq=1602 Ack=1606 Win=64128 Len=0 TSval=3718... |
| 34  | 21.312919439 | 192.168.1.100 | 192.168.1.10 | SSHv2    | 82     | Client: New Keys                                                 |



# ACL 통신 확인(서버)

## ■ Server1 ACL 정책 적용 삭제

### ➤ 정책 확인

```
root@Server1: ~  
root@Server1:~# iptables -F  
root@Server1:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Server1:~#
```

### ➤ Client2 → Server1 SSH 통신 확인

```
ubuntu@Server1: ~  
root@client2:~# ssh ubuntu@192.168.1.10  
^C  
root@client2:~# ssh ubuntu@192.168.1.10  
ssh: connect to host 192.168.1.10 port 22: Connection timed out  
root@client2:~# ssh ubuntu@192.168.1.10  
ubuntu@192.168.1.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
 * Documentation:  https://help.ubuntu.com  
 * Management:    https://landscape.canonical.com  
 * Support:       https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your  
Internet connection or proxy settings  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
Last login: Sun Nov 27 20:06:04 2022 from 192.168.1.100  
ubuntu@Server1:~$
```

```
ubuntu@client2: ~  
ubuntu@client2:~$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      0 203.230.17.10:52288     192.168.1.10:22        ESTABLISHED  
tcp6       0      0 :::22                  :::*                    LISTEN  
tcp6       0      0 :::1:631               :::*                    LISTEN  
ubuntu@client2:~$
```

# ACL 통신 확인(방화벽)

## ■ Firewall ACL 정책 적용

### ➤ 정책 확인

```
root@Firewall: ~  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

### ➤ 203.230.17.0/24 네트워크에서 Server1로 SSH접속 불가 정책 설정

```
root@Firewall: ~  
root@Firewall:~# iptables -A FORWARD -d 192.168.1.10 -p tcp --dport 22 -j DROP  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
DROP        tcp  --  anywhere             192.168.1.10          tcp dpt:ssh  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

# ACL 통신 확인(방화벽)

## ■ Client2 → Server1 통신 테스트(비정상 통신)

### ➤ SSH 통신 시도

```
ubuntu@client2: ~  
ubuntu@client2:~$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      1 203.230.17.10:43234     192.168.1.10:22        SYN_SENT  
tcp6       0      0 :::22                   :::*                     LISTEN  
tcp6       0      0 :::1:631                 :::*                     LISTEN  
ubuntu@client2:~$  
  
root@client2: ~  
root@client2:~# ssh ubuntu@192.168.1.10
```

### ➤ Firewall 에서의 패킷

```
root@Firewall: ~  
0 packets dropped by kernel  
root@Firewall:~# tcpdump -i enp0s8 dst 192.168.1.10  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s8, link-type EN10MB (Ethernet), capture size 262144 bytes  
21:05:22.680336 IP 203.230.17.10.43234 > 192.168.1.10.ssh: Flags [S], seq 2654437665, win 64240, options [mss 1460,sackOK,TS val 3583430697 ecr 0,nop,wscale 7], length 0  
21:05:23.728539 IP 203.230.17.10.43234 > 192.168.1.10.ssh: Flags [S], seq 2654437665, win 64240, options [mss 1460,sackOK,TS val 3583431745 ecr 0,nop,wscale 7], length 0  
21:05:25.747152 IP 203.230.17.10.43234 > 192.168.1.10.ssh: Flags [S], seq 2654437665, win 64240, options [mss 1460,sackOK,TS val 3583433763 ecr 0,nop,wscale 7], length 0  
21:05:30.061223 IP 203.230.17.10.43234 > 192.168.1.10.ssh: Flags [S], seq 2654437665, win 64240, options [mss 1460,sackOK,TS val 3583438078 ecr 0,nop,wscale 7], length 0  
21:05:38.219690 IP 203.230.17.10.43234 > 192.168.1.10.ssh: Flags [S], seq 2654437665, win 64240, options [mss 1460,sackOK,TS val 3583446236 ecr 0,nop,wscale 7], length 0
```

# ACL 통신 확인(방화벽)

## ■ Kali → Client1 통신 테스트

### ➤ SSH 통신 시도

```
ubuntu@client1: ~  
File Actions Edit View Help  
  
(root@kali)-[/home/kali]  
# ssh ubuntu@192.168.1.100  
The authenticity of host '192.168.1.100 (192.168.1.100)' can't be established.  
ED25519 key fingerprint is SHA256:tvaz0loOX6rwNtqpmfGz5piux5Job6DFwxWXwbUkjA.  
This key is not known by any other names  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '192.168.1.100' (ED25519) to the list of known hosts.  
ubuntu@192.168.1.100's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
  
The programs included with the Ubuntu system are free software;  
the exact distribution terms for each program are described in the  
individual files in /usr/share/doc/*/copyright.  
  
Ubuntu comes with ABSOLUTELY NO WARRANTY, to the extent permitted by  
applicable law.  
ubuntu@client1:~$  
  
kali@kali: ~  
File Actions Edit View Help  
  
(kali@kali)-[~]  
$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 203.230.17.100:55056    192.168.1.100:22      ESTABLISH  
ED  
  
(kali@kali)-[~]  
$
```

# 기본정책 변경을 이용한 ACL 통신 확인

## ■ Firewall 기본 정책 변경

### ➤ FORWARD 패킷에 대해 ACCEPT → DROP

```
root@Firewall: ~  
root@Firewall:~# iptables -P FORWARD DROP  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy DROP)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

# 방화벽 통신 확인

## ■ Client2 → Server1 통신 테스트(비정상 통신)

### ➤ SSH 통신 시도

```
ubuntu@client2: ~  
ubuntu@client2:~$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22              0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631           0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53           0.0.0.0:*               LISTEN  
tcp        0      1 203.230.17.10:37796     192.168.1.10:22        SYN_SENT  
tcp6       0      0 :::22                   :::*                     LISTEN  
tcp6       0      0 :::1:631                :::*                     LISTEN  
ubuntu@client2:~$  
  
root@client2: ~  
root@client2:~# ssh ubuntu@192.168.1.10
```

### ➤ Firewall 에서의 패킷

```
root@Firewall: ~  
root@Firewall:~#  
root@Firewall:~# tcpdump -i enp0s8 tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s8, link-type EN10MB (Ethernet), capture size 262144 bytes  
21:25:00.783344 IP 203.230.17.10.37796 > 192.168.1.10.ssh: Flags [S], seq 3803969918, win 64240, op  
tions [mss 1460,sackOK,TS val 3584608804 ecr 0,nop,wscale 7], length 0  
21:25:01.812283 IP 203.230.17.10.37796 > 192.168.1.10.ssh: Flags [S], seq 3803969918, win 64240, op  
tions [mss 1460,sackOK,TS val 3584609833 ecr 0,nop,wscale 7], length 0  
21:25:03.849014 IP 203.230.17.10.37796 > 192.168.1.10.ssh: Flags [S], seq 3803969918, win 64240, op  
tions [mss 1460,sackOK,TS val 3584611870 ecr 0,nop,wscale 7], length 0  
21:25:07.854472 IP 203.230.17.10.37796 > 192.168.1.10.ssh: Flags [S], seq 3803969918, win 64240, op  
tions [mss 1460,sackOK,TS val 3584615875 ecr 0,nop,wscale 7], length 0  
21:25:16.067681 IP 203.230.17.10.37796 > 192.168.1.10.ssh: Flags [S], seq 3803969918, win 64240, op  
tions [mss 1460,sackOK,TS val 3584624087 ecr 0,nop,wscale 7], length 0  
^C  
5 packets captured
```

# 방화벽 통신 확인

## ■ Kali → Client1 통신 테스트

### ➤ SSH 통신 시도

```
root@kali: /home/kali
File Actions Edit View Help

(root@kali)-[/home/kali]
# ssh ubuntu@192.168.1.100

kali@kali: ~
File Actions Edit View Help

(kali@kali)-[~]
$ netstat -ltan
Active Internet connections (servers and established)
Proto Recv-Q Send-Q Local Address           Foreign Address         State
tcp        0      0 203.230.17.100:43268    192.168.1.100:22        SYN_SENT
(kali@kali)-[~]
$
```

### ➤ Firewall 에서의 패킷

```
root@Firewall: ~
tcpdump -i enp0s8 tcp port 22
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s8, link-type EN10MB (Ethernet), capture size 262144 bytes
21:28:42.242936 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,
options [mss 1460,sackOK,TS val 809045421 ecr 0,nop,wscale 7], length 0
21:28:43.258411 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,
options [mss 1460,sackOK,TS val 809046436 ecr 0,nop,wscale 7], length 0
21:28:45.275026 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,
options [mss 1460,sackOK,TS val 809048452 ecr 0,nop,wscale 7], length 0
21:28:49.339224 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,
options [mss 1460,sackOK,TS val 809052516 ecr 0,nop,wscale 7], length 0
21:28:51.131470 IP 203.230.17.100.55056 > 192.168.1.100.ssh: Flags [FP.], seq 2737646097:2737646445
, ack 3874849658, win 501, options [nop,nop,TS val 809054308 ecr 2736498149], length 348
21:28:57.532046 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,
options [mss 1460,sackOK,TS val 809060708 ecr 0,nop,wscale 7], length 0
^C
6 packets captured
```

# 방화벽 통신 확인

## ■ FORWARD 정책 추가

- 192.168.1.0/24에서 외부로의 모든 통신 허용

```
root@Firewall: ~  
root@Firewall:~# iptables -A FORWARD -s 192.168.1.0/255.255.255.0 -j ACCEPT  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source      destination  
  
Chain FORWARD (policy DROP)  
target      prot opt source      destination  
ACCEPT      all  --  192.168.1.0/24  anywhere  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source      destination  
root@Firewall:~#
```



# 방화벽 통신 확인

## ■ Client1 → Client2 통신 테스트

### ➤ SSH 통신 시도

```
root@client1: ~  
root@client1:~# ssh ubuntu@203.230.17.10  
  
ubuntu@client1:~$ netstat -ltan  
Active Internet connections (servers and established)  
Proto Recv-Q Send-Q Local Address           Foreign Address         State  
tcp        0      0 0.0.0.0:22             0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.1:631          0.0.0.0:*               LISTEN  
tcp        0      0 127.0.0.53:53          0.0.0.0:*               LISTEN  
tcp        0      1 192.168.1.100:37134     203.230.17.10:22       SYN_SENT  
tcp6       0      0 :::1:631               :::*                    LISTEN  
tcp6       0      0 :::22                  :::*                    LISTEN  
ubuntu@client1:~$
```

### ➤ Firewall 에서의 패킷

```
root@Firewall: ~  
options [mss 1460,sackOK,TS val 809052516 ecr 0,nop,wscale 7], length 0  
21:28:51.131470 IP 203.230.17.100.55056 > 192.168.1.100.ssh: Flags [FP.], seq 2737646097:2737646445  
, ack 3874849658, win 501, options [nop,nop,TS val 809054308 ecr 2736498149], length 348  
21:28:57.532046 IP 203.230.17.100.56276 > 192.168.1.100.ssh: Flags [S], seq 2873335727, win 64240,  
options [mss 1460,sackOK,TS val 809060708 ecr 0,nop,wscale 7], length 0  
^C  
6 packets captured  
6 packets received by filter  
0 packets dropped by kernel  
root@Firewall:~# tcpdump -i enp0s8 tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s8, link-type EN10MB (Ethernet), capture size 262144 bytes  
21:55:51.810200 IP 192.168.1.100.55404 > 203.230.17.10.ssh: Flags [S], seq 2760707594, win 64240, o  
ptions [mss 1460,sackOK,TS val 337423724 ecr 0,nop,wscale 7], length 0  
21:55:51.810475 IP 203.230.17.10.ssh > 192.168.1.100.55404: Flags [S.], seq 1092417913, ack 276070  
7595, win 65160, options [mss 1460,sackOK,TS val 24258989 ecr 337423724,nop,wscale 7], length 0  
21:55:52.813276 IP 192.168.1.100.55404 > 203.230.17.10.ssh: Flags [S], seq 2760707594, win 64240, o  
ptions [mss 1460,sackOK,TS val 337424727 ecr 0,nop,wscale 7], length 0  
21:55:52.813504 IP 203.230.17.10.ssh > 192.168.1.100.55404: Flags [S.], seq 1092417913, ack 276070  
7595, win 65160, options [mss 1460,sackOK,TS val 24259992 ecr 337423724,nop,wscale 7], length 0
```

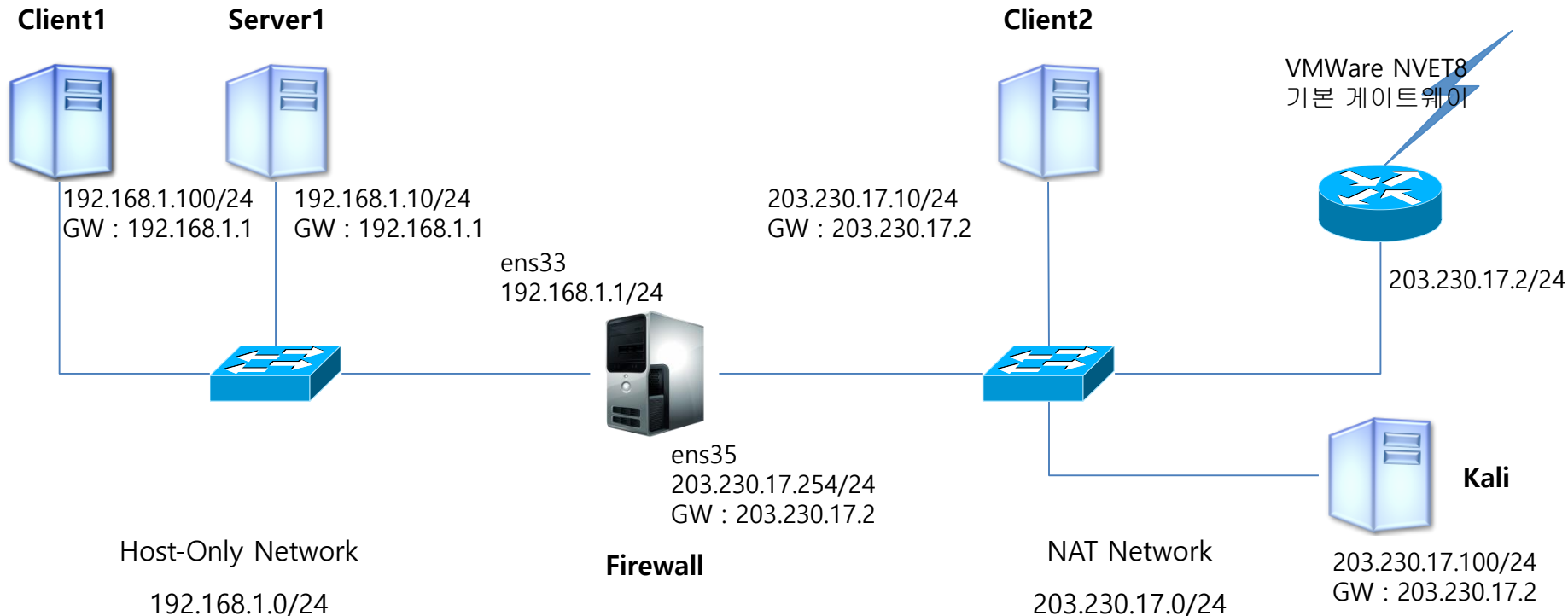
FORWARD의 기본정책을 DROP으로  
설정하였기 때문에 203.230.17.10에  
서 오는 응답 패킷이 내부로 들어가지  
못하는 현상 발생

→ 단방향 통신을 하고자하는 경우에  
만 사용하여야 함.

### 3. LINUX 방화벽

# 구성도

- 우분투 리눅스로 방화벽 테스트를 위한 구성으로 변경하여 서로 다른 네트워크간 NAT 통신을 할 수 있도록 설정하도록 합니다.



# 방화벽 설정을 위한 기본정책 초기화

## ■ Firewall 기본 정책 변경

➤ FORWARD 패킷에 대해 DROP → ACCEPT

```
root@Firewall: ~  
root@Firewall:~# iptables -F  
root@Firewall:~# iptables -P FORWARD ACCEPT  
root@Firewall:~# iptables --list  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

# 방화벽 설정을 위한 기본정책 초기화

## ■ 추가 패키지 설치

➤ apt-get install iptables-persistent netfilter-persistent

```
root@Firewall: ~  
root@Firewall:~# apt install iptables-persistent netfilter-persistent  
Reading package lists... Done  
Building dependency tree  
Reading state information... Done  
The following packages were automatically installed and are no longer required:  
  libfwupdplugin1 libxmlb1 linux-headers-5.13.0-30-generic linux-hwe-5.13-headers-5.13.0-30  
  linux-image-5.13.0-30-generic linux-modules-5.13.0-30-generic  
  linux-modules-extra-5.13.0-30-generic  
Use 'sudo apt autoremove' to remove them.  
The following NEW packages will be installed:  
  iptables-persistent netfilter-persistent  
0 upgraded, 2 newly installed, 0 to remove and 0 not upgraded.  
Need to get 13.8 kB of archives.  
After this operation, 89.1 kB of additional disk space will be used.  
Get:1 http://kr.archive.ubuntu.com/ubuntu focal-updates/universe amd64 netfilter-persistent all 1.0.14ubuntu1 [7,268 B]  
Get:2 http://kr.archive.ubuntu.com/ubuntu focal-updates/universe amd64 iptables-persistent all 1.0.14ubuntu1 [6,552 B]  
Fetched 13.8 kB in 1s (12.3 kB/s)  
Preconfiguring packages ...  
Selecting previously unselected package netfilter-persistent.  
(Reading database ... 217895 files and directories currently installed.)  
Preparing to unpack .../netfilter-persistent 1.0.14ubuntu1 all.deb ...
```

# 방화벽 설정을 위한 기본정책 초기화

## ■ iptable 조회 방법

### ➤ Filter table

- iptables -L or iptables -t filter -L

```
root@Firewall: ~  
root@Firewall:~# iptables -L  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~# iptables -t filter -L  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain FORWARD (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
root@Firewall:~#
```

# 방화벽 설정을 위한 기본정책 초기화

## ■ iptable 조회 방법

### ➤ nat table

- iptables -t nat -L

```
root@Firewall: ~  
root@Firewall:~# iptables -t nat -L  
Chain PREROUTING (policy ACCEPT)  
target      prot opt source                destination  
  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain POSTROUTING (policy ACCEPT)  
target      prot opt source                destination  
MASQUERADE  all  --  anywhere              anywhere  
root@Firewall:~#
```

# 방화벽 설정을 위한 기본정책 초기화

## ■ Tables별 사용가능 Built-in chain

### ➤ filter 테이블

|         |                              |
|---------|------------------------------|
| INPUT   | 목적지가 호스트인 패킷에 대한 필터링         |
| OUTPUT  | 출발지가 호스트인 패킷에 대한 필터링         |
| FORWARD | 호스트를 경유하는 패킷에 대한 필터링(라우터 용도) |

### ➤ nat 테이블

|             |                                                                                    |
|-------------|------------------------------------------------------------------------------------|
| PREROUTING  | target → DNAT<br>외부에서 내부로 향하는 데이터의 주소 변조<br>인터페이스에 데이터를 받자마자 변환 수행                 |
| OUTPUT      | 로컬에서 생성된 데이터를 내보내기 직전에 변환 수행                                                       |
| POSTROUTING | target → SNAT, MASQUERADE<br>내부에서 외부로 향하는 데이터의 주소 변환<br>인터페이스로 데이터가 전달되기 직전에 변환 수행 |



# Dynamic-NAT(Source NAT) 통신

- 방화벽 외부 IP를 사용하는 경우(유동 IP 사용 시) - MASQUERADE 사용
  - MASQUERADE 사용

```
root@Firewall: ~  
root@Firewall:~# iptables -t nat -A POSTROUTING -s 192.168.1.0/24 -o enp0s8 -j MASQUERADE  
root@Firewall:~# iptables -L -t nat  
Chain PREROUTING (policy ACCEPT)  
target    prot opt source                destination  
  
Chain INPUT (policy ACCEPT)  
target    prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target    prot opt source                destination  
  
Chain POSTROUTING (policy ACCEPT)  
target    prot opt source                destination  
MASQUERADE all  --  192.168.1.0/24        anywhere  
root@Firewall:~#
```

- 내부 네트워크(192.168.1.0/24)에서 외부 네트워크로 통신을 하는 경우 자동으로 방화벽의 외부 IP로 NAT되어 통신이 이루어 짐

# Dynamic-NAT(Source NAT) 통신

## ■ 방화벽 외부 IP를 사용하는 경우(유동 IP 사용 시) - MASQUERADE 사용

### ➤ Client1 → Client2 로 ping 통신

```
root@client1: ~  
root@client1:~# ping -c 2 203.230.17.10  
PING 203.230.17.10 (203.230.17.10) 56(84) bytes of data.  
64 bytes from 203.230.17.10: icmp_seq=1 ttl=63 time=0.938 ms  
64 bytes from 203.230.17.10: icmp_seq=2 ttl=63 time=0.714 ms  
  
--- 203.230.17.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1043ms  
rtt min/avg/max/mdev = 0.714/0.826/0.938/0.112 ms  
root@client1:~#
```

### ➤ TCPDUMP 화면

```
root@Firewall: ~  
root@Firewall:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:02:39.730085 IP 192.168.1.100 > 203.230.17.10: ICMP echo request, id 4, seq 1, length 64  
01:02:39.730645 IP 203.230.17.10 > 192.168.1.100: ICMP echo reply, id 4, seq 1, length 64  
01:02:40.773381 IP 192.168.1.100 > 203.230.17.10: ICMP echo request, id 4, seq 2, length 64  
01:02:40.773769 IP 203.230.17.10 > 192.168.1.100: ICMP echo reply, id 4, seq 2, length 64  
^C  
4 packets captured  
4 packets received by filter  
0 packets dropped by kernel
```

```
root@client2: ~  
root@client2:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:12:22.268135 IP 203.230.17.254 > client2: ICMP echo request, id 5, seq 1, length 64  
01:12:22.268158 IP client2 > 203.230.17.254: ICMP echo reply, id 5, seq 1, length 64
```

# Dynamic-NAT(Source NAT) 통신

- 방화벽 외부 IP를 사용하는 경우(유동 IP 사용 시) - MASQUERADE 사용
  - Client1 → Client2 로 ssh 통신

```
ubuntu@client2: ~  
root@client1:~# ssh ubuntu@203.230.17.10  
ubuntu@203.230.17.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:       https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
New release '22.04.1 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
*** System restart required ***  
Last login: Mon Nov 28 01:14:38 2022 from 203.230.17.254  
ubuntu@client2:~$
```

# Dynamic-NAT(Source NAT) 통신

- 방화벽 외부 IP를 사용하는 경우(유동 IP 사용 시) - MASQUERADE 사용
  - TCPDUMP 화면

```
root@Firewall: ~  
root@Firewall:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:14:33.791451 IP 192.168.1.100.52970 > 203.230.17.10.ssh: Flags [S], seq 2504287081, win 64240, o  
ptions [mss 1460,sackOK,TS val 1851035111 ecr 0,nop,wscale 7], length 0  
01:14:33.791782 IP 203.230.17.10.ssh > 192.168.1.100.52970: Flags [S.], seq 446602220, ack 25042870  
82, win 65160, options [mss 1460,sackOK,TS val 4180124821 ecr 1851035111,nop,wscale 7], length 0  
01:14:33.792039 IP 192.168.1.100.52970 > 203.230.17.10.ssh: Flags [.], ack 1, win 502, options [nop  
,nop,TS val 1851035111 ecr 4180124821], length 0  
01:14:33.792121 IP 192.168.1.100.52970 > 203.230.17.10.ssh: Flags [P.], seq 1:42, ack 1, win 502, o  
ptions [nop,nop,TS val 1851035112 ecr 4180124821], length 41  
01:14:33.792367 IP 203.230.17.10.ssh > 192.168.1.100.52970: Flags [.], ack 42, win 509, options [no  
p,nop,TS val 4180124821 ecr 1851035112], length 0  
01:14:33.796175 IP 203.230.17.10.ssh > 192.168.1.100.52970: Flags [P.], seq 1:42, ack 42, win 509,  
options [nop,nop,TS val 4180124825 ecr 1851035112], length 41  
01:14:33.796111 IP 192.168.1.100.52970 > 203.230.17.10.ssh: Flags [P.], seq 1:42, ack 42, win 509, options [nop,nop,TS val 1851035112 ecr 4180124825], length 41
```

```
root@client2: ~  
root@client2:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:14:33.788740 IP 203.230.17.254.52970 > client2.ssh: Flags [S], seq 2504287081, win 642  
40, options [mss 1460,sackOK,TS val 1851035111 ecr 0,nop,wscale 7], length 0  
01:14:33.788766 IP client2.ssh > 203.230.17.254.52970: Flags [S.], seq 446602220, ack 250  
4287082, win 65160, options [mss 1460,sackOK,TS val 4180124821 ecr 1851035111,nop,wscale  
7], length 0  
01:14:33.789238 IP 203.230.17.254.52970 > client2.ssh: Flags [.], ack 1, win 502, options  
[nop,nop,TS val 1851035111 ecr 4180124821], length 0  
01:14:33.789337 IP 203.230.17.254.52970 > client2.ssh: Flags [P.], seq 1:42, ack 1, win 5  
02, options [nop,nop,TS val 1851035112 ecr 4180124821], length 41  
01:14:33.789342 IP client2.ssh > 203.230.17.254.52970: Flags [.], ack 42, win 509, option  
s [nop,nop,TS val 4180124821 ecr 1851035112], length 0  
01:14:33.793157 IP client2.ssh > 203.230.17.254.52970: Flags [P.], seq 1:42, ack 42, win  
509, options [nop,nop,TS val 4180124825 ecr 1851035112], length 41
```

# Dynamic-NAT(Source NAT) 통신

## ■ 방화벽 외부 IP를 사용하는 경우(고정 IP 사용 시)

### ➤ 사용할 IP 등록

```
root@Firewall: ~  
root@Firewall:~# ifconfig enp0s8:1 203.230.17.253  
root@Firewall:~# iptables -t nat -A POSTROUTING -s 192.168.1.0/24 -o enp0s8 -j SNAT --to 203.230.17.253  
root@Firewall:~# iptables -L -t nat  
Chain PREROUTING (policy ACCEPT)  
target      prot opt source                destination  
  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain POSTROUTING (policy ACCEPT)  
target      prot opt source                destination  
SNAT        all  --  192.168.1.0/24        anywhere             to:203.230.17.253  
root@Firewall:~#
```

### ➤ 내부 네트워크(192.168.1.0/24)에서 외부 네트워크로 통신을 하는 경우 지정한 IP(203.230.17.253)로 NAT되어 통신이 이루어 짐

# Dynamic-NAT(Source NAT) 통신

## ■ 방화벽 외부 IP를 사용하는 경우(고정 IP 사용 시)

### ➤ Client1 → Client2 로 ping 통신

```
root@client1: ~  
root@client1:~# ping -c 2 203.230.17.10  
PING 203.230.17.10 (203.230.17.10) 56(84) bytes of data.  
64 bytes from 203.230.17.10: icmp_seq=1 ttl=63 time=0.938 ms  
64 bytes from 203.230.17.10: icmp_seq=2 ttl=63 time=0.714 ms  
  
--- 203.230.17.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1043ms  
rtt min/avg/max/mdev = 0.714/0.826/0.938/0.112 ms  
root@client1:~#
```

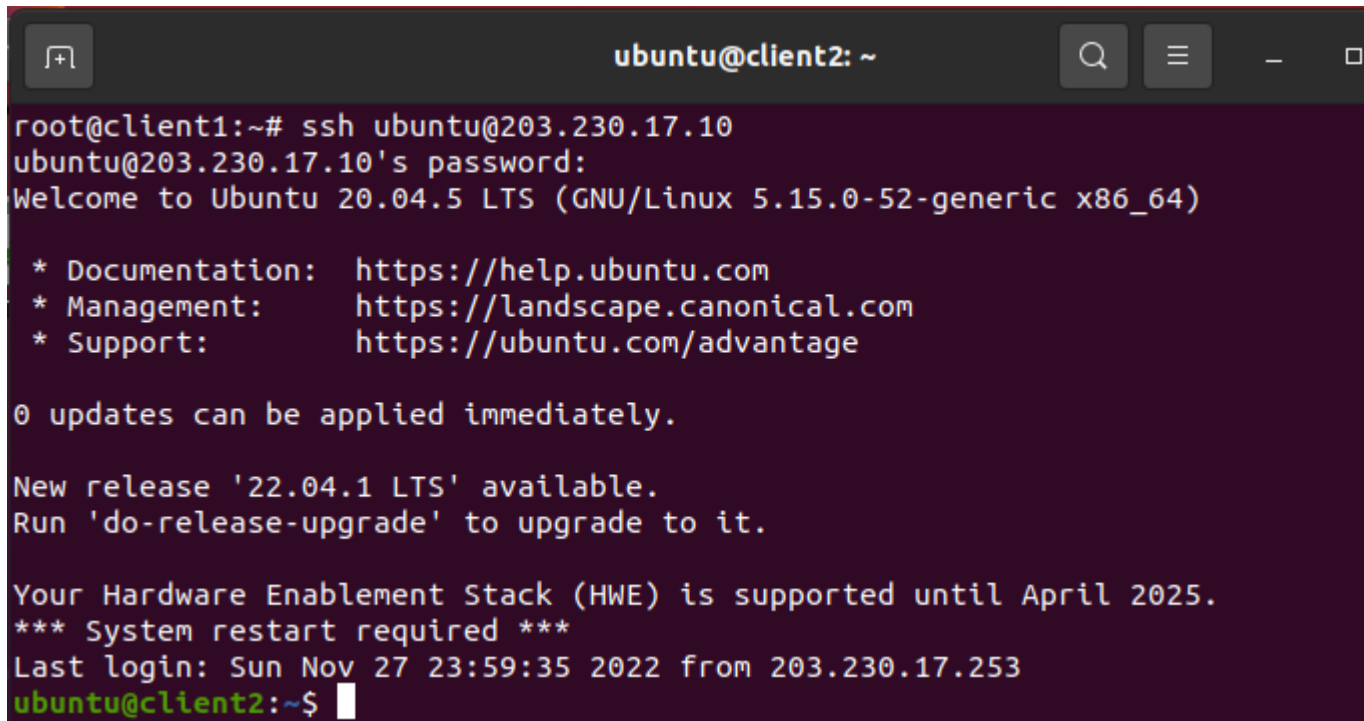
### ➤ TCPDUMP 화면

```
root@Firewall: ~  
root@Firewall:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:02:39.730085 IP 192.168.1.100 > 203.230.17.10: ICMP echo request, id 4, seq 1, length 64  
01:02:39.730645 IP 203.230.17.10 > 192.168.1.100: ICMP echo reply, id 4, seq 1, length 64  
01:02:40.773381 IP 192.168.1.100 > 203.230.17.10: ICMP echo request, id 4, seq 2, length 64  
01:02:40.773769 IP 203.230.17.10 > 192.168.1.100: ICMP echo reply, id 4, seq 2, length 64  
^C  
4 packets captured  
4 packets received by filter  
0 packets dropped by kernel
```

```
root@client2: ~  
root@client2:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:02:39.728042 ARP, Request who-has client2 tell 203.230.17.253, length 46  
01:02:39.728058 ARP, Reply client2 is-at 08:00:27:22:48:11 (oui Unknown), length 28  
01:02:39.728250 IP 203.230.17.253 > client2: ICMP echo request, id 4, seq 1, length 64  
01:02:39.728262 IP client2 > 203.230.17.253: ICMP echo reply, id 4, seq 1, length 64
```

# Dynamic-NAT(Source NAT) 통신

- 방화벽 외부 IP를 사용하는 경우(고정 IP 사용 시)
  - Client1 → Client2 로 ssh 통신



```
ubuntu@client2: ~  
root@client1:~# ssh ubuntu@203.230.17.10  
ubuntu@203.230.17.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
New release '22.04.1 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
*** System restart required ***  
Last login: Sun Nov 27 23:59:35 2022 from 203.230.17.253  
ubuntu@client2:~$
```



# Dynamic-NAT(Source NAT) 통신

## ■ 방화벽 외부 IP를 사용하는 경우(고정 IP 사용 시)

### ➤ TCPCDUMP 화면

```
root@Firewall: ~  
root@Firewall:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:06:19.791114 IP 192.168.1.100.41532 > 203.230.17.10.ssh: Flags [S], seq 2603169967, win 64240, o  
ptions [mss 1460,sackOK,TS val 1850541114 ecr 0,nop,wscale 7], length 0  
01:06:19.791483 IP 203.230.17.10.ssh > 192.168.1.100.41532: Flags [S.], seq 2774093510, ack 2603169  
968, win 65160, options [mss 1460,sackOK,TS val 1353798420 ecr 1850541114,nop,wscale 7], length 0  
01:06:19.791692 IP 192.168.1.100.41532 > 203.230.17.10.ssh: Flags [.], ack 1, win 502, options [nop  
,nop,TS val 1850541114 ecr 1353798420], length 0  
01:06:19.791880 IP 192.168.1.100.41532 > 203.230.17.10.ssh: Flags [P.], seq 1:42, ack 1, win 502, o  
ptions [nop,nop,TS val 1850541114 ecr 1353798420], length 41  
01:06:19.792062 IP 203.230.17.10.ssh > 192.168.1.100.41532: Flags [.], ack 42, win 509, options [no  
p,nop,TS val 1353798420 ecr 1850541114], length 0  
01:06:19.796812 IP 203.230.17.10.ssh > 192.168.1.100.41532: Flags [P.], seq 1:42, ack 42, win 509,  
options [nop,nop,TS val 1353798425 ecr 1850541114], length 41
```

```
root@client2: ~  
root@client2:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:06:19.788874 IP 203.230.17.253.41532 > client2.ssh: Flags [S], seq 2603169967, win 642  
40, options [mss 1460,sackOK,TS val 1850541114 ecr 0,nop,wscale 7], length 0  
01:06:19.788898 IP client2.ssh > 203.230.17.253.41532: Flags [S.], seq 2774093510, ack 26  
03169968, win 65160, options [mss 1460,sackOK,TS val 1353798420 ecr 1850541114,nop,wscale  
7], length 0  
01:06:19.789303 IP 203.230.17.253.41532 > client2.ssh: Flags [.], ack 1, win 502, options  
[nop,nop,TS val 1850541114 ecr 1353798420], length 0  
01:06:19.789488 IP 203.230.17.253.41532 > client2.ssh: Flags [P.], seq 1:42, ack 1, win 5  
02, options [nop,nop,TS val 1850541114 ecr 1353798420], length 41  
01:06:19.789494 IP client2.ssh > 203.230.17.253.41532: Flags [.], ack 42, win 509, option  
s [nop,nop,TS val 1353798420 ecr 1850541114], length 0  
01:06:19.794233 IP client2.ssh > 203.230.17.253.41532: Flags [P.], seq 1:42, ack 42, win  
509, options [nop,nop,TS val 1353798425 ecr 1850541114], length 41
```



# Static-NAT(Destination NAT) 통신

## ■ 사용할 IP 등록

```
root@Firewall: ~
root@Firewall:~# ifconfig enp0s8:1 203.230.17.250
root@Firewall:~# ip a
1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000
    link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00
    inet 127.0.0.1/8 scope host lo
        valid_lft forever preferred_lft forever
    inet6 ::1/128 scope host
        valid_lft forever preferred_lft forever
2: enp0s3: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:0e:a0:76 brd ff:ff:ff:ff:ff:ff
    inet 192.168.1.1/24 brd 192.168.1.255 scope global noprefixroute enp0s3
        valid_lft forever preferred_lft forever
    inet6 fe80::6593:f24c:6c32:e790/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
3: enp0s8: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc fq_codel state UP group default qlen 1000
    link/ether 08:00:27:10:31:76 brd ff:ff:ff:ff:ff:ff
    inet 203.230.17.254/24 brd 203.230.17.255 scope global noprefixroute enp0s8
        valid_lft forever preferred_lft forever
    inet 203.230.17.250/24 brd 203.230.17.255 scope global secondary enp0s8:1
        valid_lft forever preferred_lft forever
    inet6 fe80::6530:650d:62c:12ac/64 scope link noprefixroute
        valid_lft forever preferred_lft forever
root@Firewall:~#
```

# Static-NAT(Destination NAT) 통신

## ■ 정책 등록

```
root@Firewall: ~  
root@Firewall:~# iptables -t nat -A PREROUTING -d 203.230.17.250 -j DNAT --to 192.168.1.10  
root@Firewall:~# iptables -t nat -A POSTROUTING -s 192.168.1.10 -o enp0s8 -j SNAT --to 203.230.17.250  
root@Firewall:~# iptables -L -t nat  
Chain PREROUTING (policy ACCEPT)  
target      prot opt source                destination  
DNAT        all  --  anywhere              Firewall          to:192.168.1.10  
  
Chain INPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain OUTPUT (policy ACCEPT)  
target      prot opt source                destination  
  
Chain POSTROUTING (policy ACCEPT)  
target      prot opt source                destination  
SNAT        all  --  192.168.1.10          anywhere          to:203.230.17.250  
root@Firewall:~#
```

# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

➤ Server1 → Client2 로 ping 통신

```
root@Server1: ~  
root@Server1:~# ping -c 2 203.230.17.10  
PING 203.230.17.10 (203.230.17.10) 56(84) bytes of data.  
64 bytes from 203.230.17.10: icmp_seq=1 ttl=63 time=0.854 ms  
64 bytes from 203.230.17.10: icmp_seq=2 ttl=63 time=0.527 ms  
  
--- 203.230.17.10 ping statistics ---  
2 packets transmitted, 2 received, 0% packet loss, time 1009ms  
rtt min/avg/max/mdev = 0.527/0.690/0.854/0.163 ms  
[1]+  Done                  wireshark  
root@Server1:~#
```

➤ TCPDUMP 화면

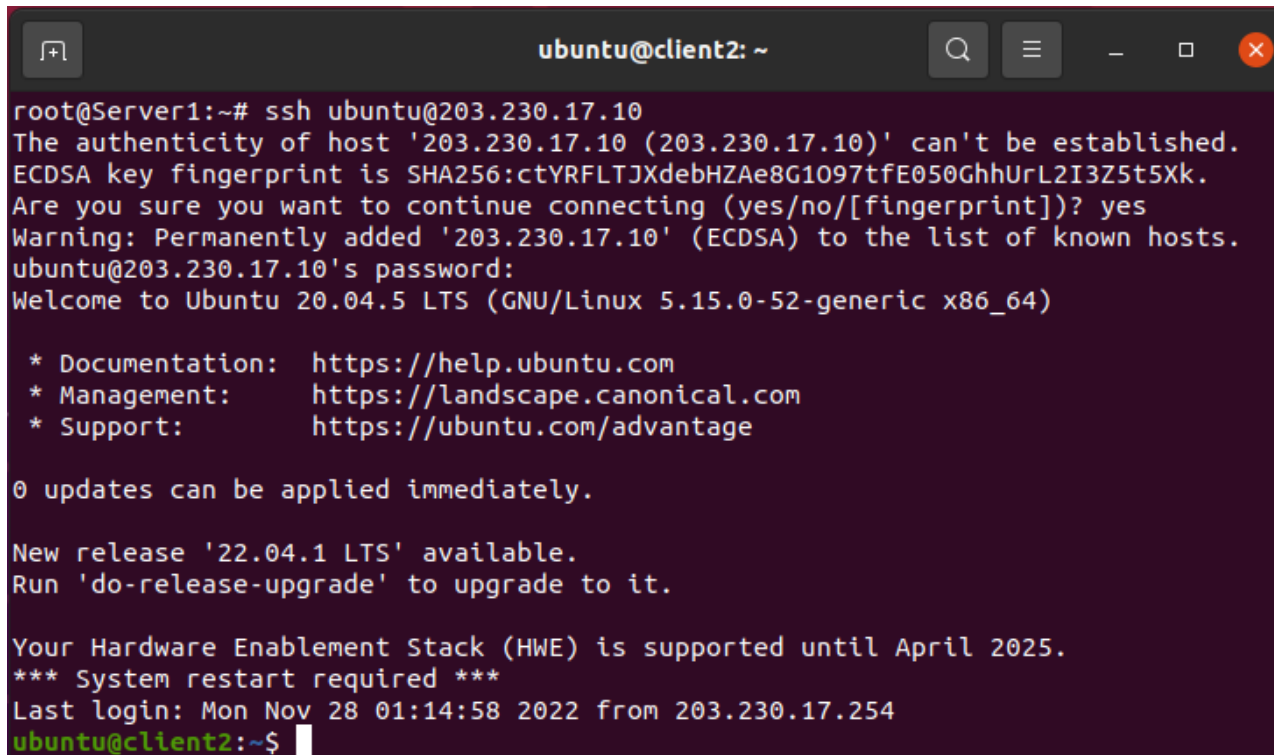
```
root@Firewall: ~  
root@Firewall:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:32:04.219240 IP 192.168.1.10 > 203.230.17.10: ICMP echo request, id 1, seq 1, length 64  
01:32:04.219792 IP 203.230.17.10 > 192.168.1.10: ICMP echo reply, id 1, seq 1, length 64  
01:32:05.228235 IP 192.168.1.10 > 203.230.17.10: ICMP echo request, id 1, seq 2, length 64  
01:32:05.228541 IP 203.230.17.10 > 192.168.1.10: ICMP echo reply, id 1, seq 2, length 64  
^C  
4 packets captured  
4 packets received by filter  
0 packets dropped by kernel  
root@Firewall:~#
```

```
root@client2: ~  
root@client2:~# tcpdump  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:32:04.213852 IP 203.230.17.250 > client2: ICMP echo request, id 1, seq 1, length 64  
01:32:04.213877 ARP, Request who-has 203.230.17.250 tell client2, length 28  
01:32:04.214102 ARP, Reply 203.230.17.250 is-at 08:00:27:10:31:76 (oui Unknown), length 46  
01:32:04.214108 IP client2 > 203.230.17.250: ICMP echo reply, id 1, seq 1, length 64
```

# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

➤ Server1 → Client2 로 ssh 통신



```
ubuntu@client2: ~  
root@Server1:~# ssh ubuntu@203.230.17.10  
The authenticity of host '203.230.17.10 (203.230.17.10)' can't be established.  
ECDSA key fingerprint is SHA256:ctYRFLTJXdebHZAe8G1097tfE050GhhUrL2I3Z5t5Xk.  
Are you sure you want to continue connecting (yes/no/[fingerprint])? yes  
Warning: Permanently added '203.230.17.10' (ECDSA) to the list of known hosts.  
ubuntu@203.230.17.10's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
New release '22.04.1 LTS' available.  
Run 'do-release-upgrade' to upgrade to it.  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
*** System restart required ***  
Last login: Mon Nov 28 01:14:58 2022 from 203.230.17.254  
ubuntu@client2:~$
```

# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

### ➤ TCPDUMP 화면

```
root@Firewall:~# tcpdump tcp port 22
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
01:35:17.329523 IP 192.168.1.10.42442 > 203.230.17.10.ssh: Flags [S], seq 405010477, win 64240, options [mss 1460,sackOK,TS val 2022350896 ecr 0,nop,wscale 7], length 0
01:35:17.329805 IP 203.230.17.10.ssh > 192.168.1.10.42442: Flags [S.], seq 827073323, ack 405010478, win 65160, options [mss 1460,sackOK,TS val 358660557 ecr 2022350896,nop,wscale 7], length 0
01:35:17.330026 IP 192.168.1.10.42442 > 203.230.17.10.ssh: Flags [.], ack 1, win 502, options [nop,nop,TS val 2022350896 ecr 358660557], length 0
01:35:17.330202 IP 192.168.1.10.42442 > 203.230.17.10.ssh: Flags [P.], seq 1:42, ack 1, win 502, options [nop,nop,TS val 2022350896 ecr 358660557], length 41
01:35:17.330462 IP 203.230.17.10.ssh > 192.168.1.10.42442: Flags [.], ack 42, win 509, options [nop,nop,TS val 358660558 ecr 2022350896], length 0
01:35:17.334203 IP 203.230.17.10.ssh > 192.168.1.10.42442: Flags [P.], seq 1:42, ack 42, win 509, options [nop,nop,TS val 358660562 ecr 2022350896], length 41
```

```
root@client2:~# tcpdump tcp port 22
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
01:35:17.323403 IP 203.230.17.250.42442 > client2.ssh: Flags [S], seq 405010477, win 64240, options [mss 1460,sackOK,TS val 2022350896 ecr 0,nop,wscale 7], length 0
01:35:17.323429 IP client2.ssh > 203.230.17.250.42442: Flags [S.], seq 827073323, ack 405010478, win 65160, options [mss 1460,sackOK,TS val 358660557 ecr 2022350896,nop,wscale 7], length 0
01:35:17.323910 IP 203.230.17.250.42442 > client2.ssh: Flags [.], ack 1, win 502, options [nop,nop,TS val 2022350896 ecr 358660557], length 0
01:35:17.324016 IP 203.230.17.250.42442 > client2.ssh: Flags [P.], seq 1:42, ack 1, win 502, options [nop,nop,TS val 2022350896 ecr 358660557], length 41
01:35:17.324022 IP client2.ssh > 203.230.17.250.42442: Flags [.], ack 42, win 509, options [nop,nop,TS val 358660558 ecr 2022350896], length 0
01:35:17.327779 IP client2.ssh > 203.230.17.250.42442: Flags [P.], seq 1:42, ack 42, win 509, options [nop,nop,TS val 358660562 ecr 2022350896], length 41
```

# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

### ➤ Kali → Server1 로 ping 통신

```
root@kali: /home/kali
File Actions Edit View Help

(root@kali) - [/home/kali]
# ping -c 2 203.230.17.250
PING 203.230.17.250 (203.230.17.250) 56(84) bytes of data.
64 bytes from 203.230.17.250: icmp_seq=1 ttl=63 time=0.741 ms
64 bytes from 203.230.17.250: icmp_seq=2 ttl=63 time=0.400 ms

--- 203.230.17.250 ping statistics ---
2 packets transmitted, 2 received, 0% packet loss, time 1015ms
rtt min/avg/max/mdev = 0.400/0.570/0.741/0.170 ms

(root@kali) - [/home/kali]
#
```

### ➤ TCPDUMP 화면

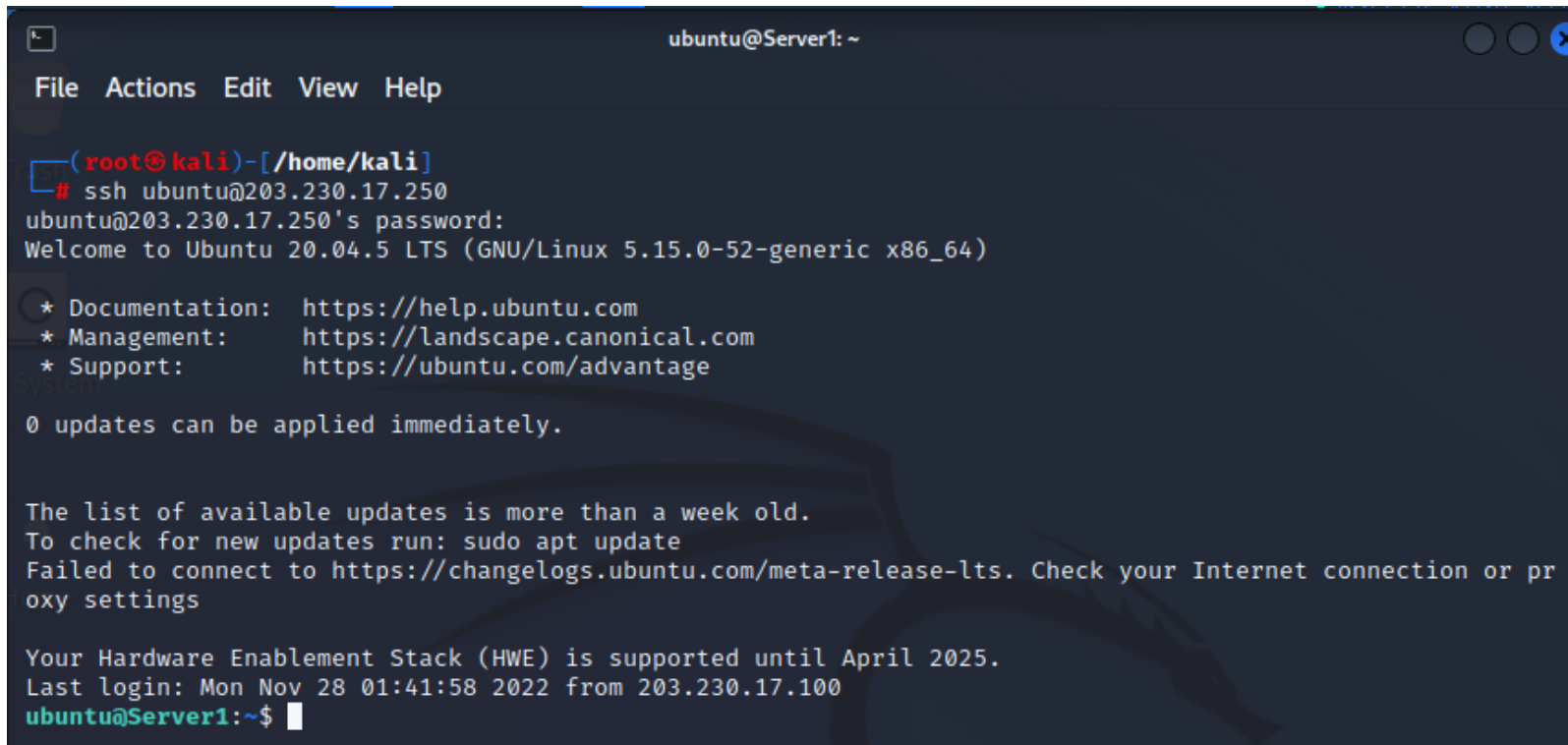
```
root@Firewall: ~
tcpdump
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
01:39:08.484522 IP 203.230.17.100 > 192.168.1.10: ICMP echo request, id 15408, seq 1, length 64
01:39:08.484785 IP 192.168.1.10 > 203.230.17.100: ICMP echo reply, id 15408, seq 1, length 64
```

```
root@Server1: ~
tcpdump
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes
01:39:08.486704 IP 203.230.17.100 > Server1: ICMP echo request, id 15408, seq 1, length 64
01:39:08.486726 IP Server1 > 203.230.17.100: ICMP echo reply, id 15408, seq 1, length 64
```

# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

➤ Kali → Server1 로 ssh 통신



```
ubuntu@Server1: ~  
File Actions Edit View Help  
  
(root@kali)-[/home/kali]  
# ssh ubuntu@203.230.17.250  
ubuntu@203.230.17.250's password:  
Welcome to Ubuntu 20.04.5 LTS (GNU/Linux 5.15.0-52-generic x86_64)  
  
* Documentation:  https://help.ubuntu.com  
* Management:    https://landscape.canonical.com  
* Support:        https://ubuntu.com/advantage  
  
0 updates can be applied immediately.  
  
The list of available updates is more than a week old.  
To check for new updates run: sudo apt update  
Failed to connect to https://changelogs.ubuntu.com/meta-release-lts. Check your Internet connection or proxy settings  
  
Your Hardware Enablement Stack (HWE) is supported until April 2025.  
Last login: Mon Nov 28 01:41:58 2022 from 203.230.17.100  
ubuntu@Server1:~$
```



# Static-NAT(Destination NAT) 통신

## ■ 통신 테스트

```
root@Firewall: ~  
root@Firewall:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:42:45.879826 IP 203.230.17.100.36296 > 192.168.1.10.ssh: Flags [S], seq 3601970733, win 64240, options [mss 1460,sackOK,TS val 2062897163 ecr 0,nop,wscale 7], length 0  
01:42:45.880241 IP 192.168.1.10.ssh > 203.230.17.100.36296: Flags [S.], seq 640088662, ack 3601970734, win 65160, options [mss 1460,sackOK,TS val 662883291 ecr 2062897163,nop,wscale 7], length 0  
01:42:45.880625 IP 203.230.17.100.36296 > 192.168.1.10.ssh: Flags [.], ack 1, win 502, options [nop,nop,TS val 2062897164 ecr 662883291], length 0  
01:42:45.880658 IP 203.230.17.100.36296 > 192.168.1.10.ssh: Flags [P.], seq 1:36, ack 1, win 502, options [nop,nop,TS val 2062897164 ecr 662883291], length 35  
01:42:45.880839 IP 192.168.1.10.ssh > 203.230.17.100.36296: Flags [.], ack 36, win 509, options [nop,nop,TS val 662883292 ecr 2062897164], length 0  
01:42:45.884574 IP 192.168.1.10.ssh > 203.230.17.100.36296: Flags [P.], seq 1:42, ack 36, win 509, options [nop,nop,TS val 662883296 ecr 2062897164], length 41
```

```
root@Server1: ~  
root@Server1:~# tcpdump tcp port 22  
tcpdump: verbose output suppressed, use -v or -vv for full protocol decode  
listening on enp0s3, link-type EN10MB (Ethernet), capture size 262144 bytes  
01:42:45.878562 IP 203.230.17.100.36296 > Server1.ssh: Flags [S], seq 3601970733, win 64240, options [mss 1460,sackOK,TS val 2062897163 ecr 0,nop,wscale 7], length 0  
01:42:45.878590 IP Server1.ssh > 203.230.17.100.36296: Flags [S.], seq 640088662, ack 3601970734, win 65160, options [mss 1460,sackOK,TS val 2062897163 ecr 2062897163,nop,wscale 7], length 0  
01:42:45.879289 IP 203.230.17.100.36296 > Server1.ssh: Flags [.], ack 1, win 502, options [nop,nop,TS val 2062897164 ecr 662883291], length 0  
01:42:45.879290 IP 203.230.17.100.36296 > Server1.ssh: Flags [P.], seq 1:36, ack 1, win 502, options [nop,nop,TS val 2062897164 ecr 662883291], length 35  
01:42:45.879309 IP Server1.ssh > 203.230.17.100.36296: Flags [.], ack 36, win 509, options [nop,nop,TS val 662883292 ecr 2062897164], length 0  
01:42:45.882983 IP Server1.ssh > 203.230.17.100.36296: Flags [P.], seq 1:42, ack 36, win 509, options [nop,nop,TS val 662883296 ecr 2062897164], length 41
```



# 4. LINUX 확장

# iptables를 이용한 NAT(Network Address Translation) 구현

## ■ NAT 테이블에서 사용하는 체인

- PREROUTING : 외부에서 내부로 보낼때 변환하는 체인  
(DNAT - 하나의 공인 IP주소로 다수의 서버를 운영하는 경우 사용)
- POSTROUTING : 내부에서 외부로 보낼때 변환하는 체인  
(SNAT - 공인 IP주소 하나로 다수의 컴퓨터가 인터넷 접속 가능)

## ■ 사용 예

- # iptables -t nat -A PREROUTING -p tcp -d 203.247.50.100 --dport 80 -j DNAT --to 192.168.0.4:80  
-> 하나의 공인 IP 주소를 이용해 다수의 서버를 연결하여 구성하고, 외부로부터 들어오는 웹 서비스 요청을 내부의 사설 IP 주소가 192.168.0.4인 웹 서버로 전달한다
- # iptables -t nat -A POSTROUTING -o eth0 -j SNAT --to 203.247.40.100  
-> 하나의 공인 IP를 공유하여 다수의 컴퓨터를 사용하는 환경을 구성하고, 해당 시스템에는 이더넷 카드가 두 개 장착되어 있는데 첫번째 이더넷 카드에서 나가는 패킷에 대해 공인 IP주소인 203.247.40.100을 할당한다

# DoS 공격 차단 시나리오

- 일반적으로 1초 동안 80번 포트에 똑같은 IP로 10번 이상의 SYN이 들어오는 경우는 적음
- 따라서 10번 이상의 SYN이 들어오는 경우 이를 공격 시도로 판단
- 10번 이상의 패킷이 반복적으로 들어오는 경우 DROP 할 수 있도록 처리
- 이 때는 다음과 같은 설정 가능(SSL이 적용된 서비스인 경우 443으로 설정)
  - 1초에 10번 이상 접근 저장 목적의 리스트 생성
    - iptables -A INPUT -p tcp --dport 80 -m recent --set --name HTTP\_Flood
  - 1초에 10번 이상 접근 로깅
    - iptables -A INPUT -p tcp --dport 80 -m recent --update --seconds 1 --hitcount 10 --name HTTP\_Flood -j LOG --log-prefix "[Attack Detected]"
  - 1초에 10번 이상 접근 차단
    - iptables -A INPUT -p tcp --dport 80 -m recent --update --seconds 1 --hitcount 10 --name HTTP\_Flood -j DROP
  - 각 IP를 가지는 사용자가 1초에 10번 이상 서버에 접근했을 때 해당 IP를 차단
  - 차단된 IP에 대한 정보는 로그로 남게 되며 마지막으로 차단된 시간으로부터 1초 뒤 다시 접속

# DoS 공격 차단 시나리오

## ■ 저장된 로그 확인

- 우분투(Ubuntu)에서 기본적으로 iptables 관련 로그는 /var/log/kern.log의 위치에 저장
- 일반적으로 다음과 같은 명령어로 로그에 기록되어 있는 공격 시도 패킷 확인
  - `grep -r "Attack Detected" /var/log/kern.log`

# DoS 공격 차단 예시

## ➤ Block Invalid Packets

- `iptables -t mangle -A PREROUTING -m conntrack --ctstate INVALID -j DROP`

## ➤ Block New Packets That Are Not SYN

- `iptables -t mangle -A PREROUTING -p tcp ! --syn -m conntrack --ctstate NEW -j DROP`

## ➤ Block Uncommon MSS Values

- `iptables -t mangle -A PREROUTING -p tcp -m conntrack --ctstate NEW -m tcpmss ! --mss 536:65535 -j DROP`

## ➤ Block Packets With Bogus TCP Flags

- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags FIN,SYN,RST,PSH,ACK,URG NONE -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags FIN,SYN FIN,SYN -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags SYN,RST SYN,RST -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags FIN,RST FIN,RST -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags FIN,ACK FIN -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags ACK,URG URG -j DROP`
- `iptables -t mangle -A PREROUTING -p tcp --tcp-flags ACK,FIN FIN -j DROP`

# DoS 공격 차단 예시

## ➤ Block Packets From Private Subnets (Spoofing)

- iptables -t mangle -A PREROUTING -s 224.0.0.0/3 -j DROP
- iptables -t mangle -A PREROUTING -s 169.254.0.0/16 -j DROP
- iptables -t mangle -A PREROUTING -s 172.16.0.0/12 -j DROP
- iptables -t mangle -A PREROUTING -s 192.0.2.0/24 -j DROP
- iptables -t mangle -A PREROUTING -s 192.168.0.0/16 -j DROP
- iptables -t mangle -A PREROUTING -s 10.0.0.0/8 -j DROP
- iptables -t mangle -A PREROUTING -s 0.0.0.0/8 -j DROP
- iptables -t mangle -A PREROUTING -s 240.0.0.0/5 -j DROP
- iptables -t mangle -A PREROUTING -s 127.0.0.0/8 ! -i lo -j DROP

## ➤ Drop ICMP (you usually don't need this protocol)

- iptables -t mangle -A PREROUTING -p icmp -j DROP

## ➤ Drop fragments in all chains

- iptables -t mangle -A PREROUTING -f -j DROP

## ➤ Limit connections per source IP

- iptables -A INPUT -p tcp -m connlimit --connlimit-above 111 -j REJECT --reject-with tcp-reset

# DoS 공격 차단 예시

## ➤ Limit RST packets

- iptables -A INPUT -p tcp --tcp-flags RST RST -m limit --limit 2/s --limit-burst 2 -j ACCEPT
- iptables -A INPUT -p tcp --tcp-flags RST RST -j DROP

## ➤ Limit new TCP connections per second per source IP

- iptables -A INPUT -p tcp -m conntrack --ctstate NEW -m limit --limit 60/s --limit-burst 20 -j ACCEPT
- iptables -A INPUT -p tcp -m conntrack --ctstate NEW -j DROP

## ➤ Protection against port scanning

- iptables -N port-scanning
- iptables -A port-scanning -p tcp --tcp-flags SYN,ACK,FIN,RST RST -m limit --limit 1/s --limit-burst 2 -j RETURN
- iptables -A port-scanning -j DROP

End of Document